Disparate Access to Credit: Investigating North Carolina's Residential Mortgage Market 2018-2021

A White Paper Report from the Research, Policy, and Impact Center

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BACKGROUND

Racial segregation in housing refers to the systematic practice of separating populations based on race or ethnicity into different geographical areas, especially in terms of residential housing. This practice has deep roots in the history of the United States, involving a complex interplay of social, economic, and political factors and has contributed to the extreme wealth gap observed between White and Black households.

Several mechanisms drove racial segregation including explicitly discriminatory practices in city planning and banking. Redlining was a notorious practice that began in the 1930s, and involved the federal government-backed Home Owners' Loan Corporation (HOLC) drawing color-coded maps of cities, marking neighborhoods with non-white residents, especially African Americans, in red to denote them as hazardous areas for mortgage lending. Banks and insurance companies used these maps to justify denying loans and insurance to residents in these areas or charging them exorbitantly high rates. These practices were pervasive and have had enduring effects on American cities.

The Fair Housing Act of 1968 was established to address discriminatory practices and disparities in the housing market. It was a landmark legislation in the Civil Rights Movement, coming in the wake of the assassination of Dr. Martin Luther King Jr. The Act prohibited discrimination concerning the sale, rental, or financing of housing based on race, color, religion, sex, or national origin. It also mandated the Department of Housing and Urban Development (HUD) to actively promote racial integration in housing.

The Fair Housing Act of 1968 aimed to reverse decades of institutionalized housing discrimination and promote racial integration. However, the effectiveness of the Act has been a subject of debate among scholars, with many arguing that enforcement has been inadequate, and the impacts of past discriminatory practices continue to persist, resulting in enduring racial segregation in American cities.

ACCESS TO CREDIT

Access to credit is an important factor for economic growth and development and individual financial well-being. It plays an imperative role in facilitating homeownership, business investments, and overall economic growth. However,

disparities in credit access have been a persistent concern, particularly in relation to sex and race. Understanding the factors that influence credit access and identifying potential disparities is essential for promoting financial inclusion and addressing systemic inequalities.

Despite the passage of the Fair Housing Act in 1968, evidence suggested that discriminatory practices were still prevalent in banking and lending in the 1970s. These institutions refused to extend mortgages or offered less favorable loan terms in certain neighborhoods, typically those predominantly inhabited by racial minorities. Women were also less likely to receive a loan. A study by Card (1980) asserts that in the mid-1970s homeownership by women was usually associated with marriage.[1] During this period, the housing market and lending practices were tailored to cater to married couples. It is safe to assume that societal norms and expectations regarding homeownership and sex roles have evolved but the question is by how much?

Activists, researchers, and policy makers sought better data to understand the scope of these problems and to enforce anti-discrimination laws more effectively. However, such lending data were not available at the time. The Home Mortgage Disclosure Act (HMDA) was signed into law in the United States in 1975 against the backdrop of considerable civil rights activism and growing concerns about racial discrimination in housing and credit markets. It was part of a broader effort to address systemic racism and socioeconomic disparities in the United States.

HMDA mandated that lending institutions report mortgage lending activities through a standard reporting form, known as the Loan Application Register (LAR). This provision allowed the public, policymakers, and researchers to evaluate whether lending institutions were serving the housing needs of their communities and whether there is any evidence of discriminatory lending practices. Each entry in the LAR provides details such as the borrower's race, sex, and income, the loan amount, the type of loan, and whether the application was accepted or denied. This data is crucial in discerning patterns that might suggest unequal access to residential mortgages.

Academic studies using LAR data have extensively examined the existence and impact of disparate lending practices. Scholars have also used LAR data to analyze the outcomes of policy changes, such as the Fair Housing Amendments Act of 1988,

[1] Card, E. (1980). Women, Housing Access, and Mortgage Credit. Signs, 5(3), S215–S219. JSTOR.

or the effects of the subprime lending crisis on different population segments. Through a temporal analysis of this data, they have shed light on the persistent housing discrimination faced by marginalized groups. Additionally, the LAR data are also used to identify discriminatory practices like predatory lending, where loans with unfair and abusive loan terms are targeted towards vulnerable borrowers. High-interest rates, excessive fees, penalties, deceptive practices and marketing tactics, can be traced by linking patterns in the LAR data with demographic and geographic variables. The Loan Application Register (LAR) data provides an essential tool to assess the fulfillment of the objectives of the Home Mortgage Disclosure Act (HMDA). This data allows a rigorous evaluation of the residential mortgage landscape, revealing disparities in mortgage access, discriminatory lending practices, and the effectiveness of housing policies.

Recent studies utilizing the HMDA dataset have shed light on disparities in credit access based on sex and race. These studies have revealed significant variations in loan approval rates, loan amounts, and interest rates across different demographic groups. Such disparities have raised concerns about potential discrimination and have prompted calls for policy interventions to promote equal access to credit. For example, a paper by (Zhang & Willen, 2021) found that minority applicants especially Blacks and Hispanics pay higher interest rates than White applicants.[2]

The results of this research serve as a resource for policymakers and other stakeholders to guide policy decisions and initiatives that promote fair lending practices and reduce inequalities in access to credit. Financial institutions can also leverage these findings to refine their lending practices and develop strategies that ensure fair and unbiased treatment of borrowers and/or applicants from different demographic backgrounds to ensure an equitable financial system and achieve the goals of financial inclusivity. Considering this, conducting a comprehensive analysis of the HMDA dataset will provide valuable insights into the dynamics of access to credit and contribute to ongoing discussions surrounding financial inclusion and equality.

[2] Zhang, D. H., & Willen, P. S. (2021). Do Lenders Still Discriminate? A Robust Approach for Assessing Differences in Menus. National Bureau of Economic Research.

METHODOOLGY

This study employs data from the Home Mortgage Disclosure Act (HMDA) to test if lenders are serving the housing needs of all borrowers. It provides comprehensive information on loan applications, loan characteristics, and the demographics of the individuals who apply. Our analysis focuses on the period from 2018 to 2021 to investigate the relationship between an individual's sex and/or race, and their access to credit.

To conduct this analysis, a subset of the HMDA dataset focusing on mortgage loan applicants in North Carolina was extracted. The selection process filtered out non-relevant data, narrowing the scope to individuals who applied for mortgage loans within the specified timeframe and geography. Variables of interest in this study comprise sex, race, loan application outcomes (approval or denial), and loan characteristics (such as loan amount and purpose).

The primary objective of this report is to study credit access patterns and explore the influence of sex and race on an individual's likelihood of obtaining credit. Descriptive statistics are used to provide an overview of credit access patterns within the dataset and summarize the distribution of variables such as loan amount, loan purpose, and loan outcomes across different demographic groups.

Binary logistic regression is used to further examine the association between sex, race, and access to credit, shedding light on potential disparities in credit access across different demographic groups. This analysis is used to understand the relationship between this binary outcome and a set of independent (or predictor) variables, which can be either continuous (like age or income) or categorical (like sex or race). Logistic regression works by estimating the probability of a particular outcome occurring, such as the probability of receiving a loan (coded as 1) or not receiving a loan (coded as 0), given certain predictor variables, such as the race or sex of the applicant.

This modeling approach estimates the likelihood of loan approval while considering (or holding constant) other relevant factors, including loan size, creditworthiness, and income. By including these covariates, the logistic regression model allows for isolating the specific effects of sex and race on credit access.

LIMITATIONS

In considering the findings of this study, it is important to acknowledge several limitations. Firstly, the analysis relies on the HMDA dataset, which may have limitations in capturing the entirety of credit access and potential factors that influence loan outcomes. The dataset may not include certain variables that could be significant in understanding credit access. Factors such as employment history, financial education and literacy, debt-to-income ratio, and previous credit history are not explicitly recorded. Consequently, the absence of these variables could restrict the ability to fully grasp the complexity of credit access determinants.

Furthermore, like any large-scale dataset, the HMDA data may suffer from inaccuracies or incomplete information. Mistakes in data entry or reporting by financial institutions can introduce discrepancies or result in missing values, which in turn may introduce biases or affect the reliability of the study's findings. To mitigate these concerns, diligent efforts have been made to meticulously clean and validate the dataset.

DESCRIPTIVE ANALYSIS OF HMDA DATA

The purpose of the descriptive analysis performed was to provide valuable insights into the variables present in the (Home Mortgage Disclosure Act) HMDA dataset spanning the years 2018 to 2021. The analysis begins with a comprehensive exploration of each variable individually, examining their distributions, central tendencies, and levels of variability for the entire state of North Carolina. Through these investigations we aim to gain a deeper understanding of the properties and variations of each variable

Additionally, the analysis goes beyond the isolated study of variables. It includes comparisons between variables to identify potential relationships and patterns that may be present in the dataset. By examining these relationships, we are able to obtain some valuable insight into the influences and relationships between the variables.

Also, the analysis then focuses on seven major cities in the state of North Carolina: Raleigh, Durham, Greensboro/ High Point, Winston-Salem, Charlotte, Asheville, and Wilmington. This approach allows us to explore any differences or similarities in different city contexts. By comparing the data of these specific locations, we get

the opportunity to spot different trends or peculiarities that may be present in each city.

LOAN APPLICATION CHARACTERISTICS

The following section describes the loan characteristics for applications filed in the state of North Carolina as well as the general characteristics of the lending institutions.

Table 1 – Type of Loan (North Carolina)

Loan Type	2018 Overall (%)	2019 %	2020 %	2021 %	Total Period
Conventional	73.69	73.04	74.93	77.56	2,171,896
FHA	12.63	12.04	9.34	8.71	296,718
VA	11.59	13.43	14.16	12.46	375,932
FSA/RHS	2.09	1.48	1.57	1.18	43875
Total N	506,809	596,174	860,520	924,918	2,888,421

Type of Loan

The analysis below provides valuable insights into the distribution of loan types across the years 2018 to 2021. The data reveals interesting patterns and trends that shed light on the dynamics of lending in the examined period. Conventional loans displayed consistent dominance throughout the years accounted for 75.19% of the total 2,888,421 loans applied for over the 4-year period suggesting a sustained demand or preference for this loan type in North Carolina. In 2018, this loan type accounted for (373,492) 73.69% of all loans.

Federal Housing Administration (FHA) loans accounted for (296,718) 10.27% of all loans in this period. FHA demonstrated a consistent reduction in terms of its percentage despite the increased number of FHA loans applied over the years.

Department of Veteran's Affairs (VA) loans, although initially accounting for (58,726) 11.59% of all loans in 2018, experienced a substantial increase over time. The count rose to 375,932 loans (13.02% of all loans) in the entire period. This notable upward trend suggests a growing demand for VA loans.

Finally, Rural Housing Loans (RHS) or Farm Service Agency (FSA) loans exhibited relatively smaller numbers compared to the other categories. In 2018, they represented (10,600) 2.09% of all loans. The count remained relatively low in subsequent years, reaching 43,875 loans overall (1.52% of all loans). (See Table 1)

A similar analysis was run for the aforementioned individual cities and the results are shown in the tables below. Conventional loans are the most common type applied for across all cities ranging from over 50% to about 87% and show a generally increasing trend over the year, except for Wilmington whose trend is not obvious. The percentages for Farm Service Agency (FSA) or Rural Housing Service (RHS) loans are relatively small compared to conventional, FHA, and VA loans indicating they were the least sought out of the loans for cities as they typically require residence in areas that are designated as rural. The percentages range from 0.45% to 2.41%. The tables in the appendices present the findings for each city.

Table 2 - Property Type (North Carolina)

Property Type	2018 %	2019 %	2020 %	2021 %	Total Period
Multifamily: Manufactured	0.02	0.02	0.01	0.01	425
Multifamily: Site-Built	0.24	0.20	0.15	0.15	5,053
Single Family (1-4 Units): Manufactured	5.94	5.55	4.32	4.62	143,125
Single Family (1-4 Units): Site-Built	93.80	94.23	95.52	95.22	2,739,818
Total N	506,809	596,174	860,520	924,918	2,888,421

Property Type

For the period under analysis, site-built single-family dwellings represented the most observed dwellings accounting for (2,739,818) 94.86% of all observed dwellings, while those manufactured accounted for (143,125) 4.96% of dwellings observed.

Multifamily Dwellings exhibited a consistently low count throughout the years for both site-built and manufactured. In 2018, there were only 95 and 1,195 instances for site-built and manufactured respectively, which accounted for a mere 0.02% and 0.24% of all observed dwellings. The count slightly increased in subsequent years but remained relatively insignificant, reaching a total of 425 and 5,053 instances respectively across the entire period. (See Table 2 below).

We observe a similar analysis for the individual cities. Majority of observed dwellings were site-built single-family dwellings for all cities and Multifamily dwellings that were manufactured accounted for about 0.02% across all observed

cities. The tables below provide a comprehensive and detailed breakdown of the analysis results.

Table 3 - Purpose of the Loan (North Carolina)

Purpose of Loan	2018 %	2019 %	2020 %	2021 %	Total Period
Home purchase	54.49	49.21	36.97	36.94	1,229,365
Home improvement	6.99	6.37	3.93	4.29	146,965
Other purpose	8.58	7.28	4.15	4.22	161,557
Not applicable	1.57	1.30	1.46	0.40	32,009
Refinancing	14.60	22.13	41.97	37.15	910,694
Cash-out Refinancing	13.76	13.70	11.52	17.00	407,831
Total N	506,809	596,174	860,520	924,918	2,888,421

Purpose of Loan

Understanding the loan purposes is crucial for assessing the nature of lending activities and identifying any shifts or trends in borrower preferences.

In 2018, (276,182) 54.49% of all loans were obtained for the purposes of home purchase. Although the share for home purchase has slightly decreased in subsequent years, it remained the dominant loan purpose throughout the period, reaching a total count of 1,229,365 loans (42.56% of all loans). For the entire period, 146,965 loans (5.09%) were applied for the purposes of home improvement. In 2018, it accounted for 6.99% of all loans, with a count of 35,447. Refinancing accounted for 910,694 (31.53%) of all loans and cash-out refinancing accounted for 407,831 loans (14.12% of all loans) in the entire period. (See Table 3)

In the individual cities (see Appendices), loans obtained for the purpose of home purchase consistently hold the highest percentage. We note that there has been a noticeable decreasing trend in these types of loans over the years. On the other hand, loans taken for refinancing exhibit a generally increasing trend and rank as the second-highest category across all cities. It is worth mentioning that both loans for home improvement and loans for other purposes each accounted for less than 10% in every year and city under consideration.

Table 4 - Owner Occupancy (North Carolina)

Occupancy Type	2018 %	2019 %	2020 %	2021 %	Total Period
Principal residence	90.28	90.88	91.00	89.77	2,612,763
Secondary residence	4.17	3.87	4.25	4.14	119,034
Investment Property	5.45	5.25	4.75	6.10	156,624
Total N	506,809	596,174	860,520	924,918	2,888,421

Occupancy Type

Primary residence accounted for the majority of the observations in each year. In 2018, it constituted 90.29% (457,606) of the total observations. It accounted for 90.88% in 2019, 91% in 2020 and 89.77% in 2021. The proportion of primary residence remained consistently high throughout the analyzed period, with a total count of 2,612,763 observations (90.46% of the total).

In 2018, secondary residence accounted for 4.17% of the total observations, with a count of 21,117. The overall count for secondary residence was 119,034 observations (4.12% of the total) during the entire period. For the entire period, less than 6% (156,624) of applicants applied for loans for investment property. In 2018, it accounted for 5.54% of the total observations, with a count of 28,086.

We observe a strong preference for homeownership as the primary occupancy type across the cities analyzed with percentages varying from 77% to about 97%. Secondary residence loans, representing properties used as vacation homes or secondary dwellings, accounted for the smallest proportion of loans in each city. Investment property loans, which are acquired for rental or investment purposes, constituted a smaller portion compared to principal residence loans in every city varying from about 4% to about 7%. The tables below show a detailed breakdown of the data analysis.

Table 5 - Loan Amount (North Carolina)

Loan Amount	2018 ($)	2019 ($)	2020 ($)	2021 ($)	Total Period
Mean	196,837	217,091.2	236,654.7	248,248.8	229,342.9
Median	155,000	175,000	205,000	205,000	185,000
Minimum	5,000	5,000	5,000	5,000	5,000
Maximum	360,000,000	900,000,000	140,000,000	366,000,000	900,000,000
N	506,809	596,174	860,520	924,918	2,888,421

Amount of Loan

The loan amount requested on applications ranged from a minimum of $5,000 to a maximum of $900,000,000 over the period (see Table 5 below). From 2018 to 2021, the mean loan amount increased gradually from $196,837 to $248,248.8, with an overall average of $229,342.9. In 2018, the median loan amount was $155,000, which increased to $175,000 in 2019. In 2020, the median amount further rose to $205,000, and it remained the same in 2021. The overall median loan amount across the entire period was $185,000. The minimum loan amount remained constant at $5,000 throughout the four-year period.

The maximum loan amount, on the other hand, varied significantly across the years. In 2018, the maximum loan amount reached $360,000,000, which increased to $900,000,000 in 2019. In 2020, then decreased to $140,000,000, and in 2021, it slightly rose to $366,000,000. The tables in the Appendices show a similar analysis for the individual cities.

Table 6 - Pre-approval (North Carolina)

Pre-approval Status	2018 %	2019 %	2020 %	2021 %	Total Period
Pre-approval requested	1.76	1.68	0.87	0.98	35,474
Pre-approval not requested	98.24	98.32	99.13	99.02	2,852,947
N	506,809	596,174	860,520	924,918	2,888,421

Pre-approval

In 2018, out of a total of 506,809 observations, 8,931 loans (1.76% of the total) requested preapproval while 497,878 loans (98.24% of the total) did not. Similarly, in 2019, 10,010 loans (1.68% of the total) requested preapproval, and 586,164 loans (98.32% of the total) did not. In 2020 and 2021, less than a percent requested preapproval. (See tables below)

Table 7 - Crosstabulation of Approval Vs. Pre-Approval Request (North Carolina)

Pre-approval Status	2018 %	2019 %	2020 %	2021 %	Total Period
Pre-approval requested					35,474
Loan Approved	8,319 (93.15)	9,508 (94.99)	7,034 (94.31)	7,892 (86.96)	32,753 (92.33)
Not Approved	612 (6.85)	502 (5.01)	424 (5.69)	1,183 (13.04)	2,721 (7.67)
Pre-approval not requested					2,852,947
Loan Approved	328,387 (65.96)	395,680 (67.50)	589,739 (69.13)	637,749 (69.64)	1,951,555 (68.40)
Not Approved	169,491 (34.04)	190,484 (32.50)	263,323 (30.87)	278,094 (30.36)	901,392 (31.60)
N	506,809	596,174	860,520	924,918	2,888,421

Pre-Approval by Approve Loan

The table above provides a crosstabulation of loan approval status against pre-approval request for the entire period. Of those requesting preapproval for the period,32,753 were approved compared to the 1,951,555 of those who did not request preapproval. Loans that requested preapproval had a significantly higher approval rate of 92.33%, compared to the approval rate of 68.40% for loans that did not request preapproval. This suggests that requesting preapproval for a loan in North Carolina is associated with a higher likelihood of approval highlighting the importance of considering preapproval as a strategy to increase the chances of loan approval for borrowers.

Loan Action

In accordance with the reporting convention of other Fair Housing Agencies (1997), the loan Application register data variable "Loan Action" was originally read in as "Loan originated" (56.50%), "Application approved but not accepted" (2.25%), or "Purchased loan" (9.96 %) were recoded as "Approved" for this analysis. About two-thirds of the loan applications were approved (68.7% or 1,984,308 applications). Applications coded as "Application denied" (12.99%), "Application withdrawn by applicant" (12.68 %), "File closed for incompleteness" (5.53%), "Preapproval request denied" (0.04%), or "Preapproval request approved but not accepted" (0.06%) were recoded as "Not Approved (Denied, Withdrawn, or Incomplete)" and represented about a third of applications (31.3% or 904,113 applications) (see Tables below).

Table 8 - Loan Action (North Carolina)

Loan Action	2018 %	2019 %	2020 %	2021 %	Total Period
Loan originated	52.87	54.32	57.57	58.88	1,631,814
Application approved but not accepted	2.03	2.10	2.31	2.40	64,879
Application denied	16.68	14.61	11.60	11.23	375,343
Application withdrawn by applicant	12.46	12.54	13.32	12.29	366,207
File closed for incompleteness	4.30	4.80	5.68	6.55	159,842
Purchased loan	11.54	11.54	9.47	8.52	287,615
Preapproval request denied	0.07	0.04	0.03	0.03	1,125
Preapproval request approved but not accepted	0.05	0.05	0.02	0.10	1,596
Total N	506,809	596,174	860,520	924,918	2,888,421

Table 9 - Loan Approval (North Carolina)

Loan Action	2018 %	2019 %	2020 %	2021 %	Total Period
Not approved (Denied, Withdrawn, or Incomplete)	33.56	32.04	30.65	30.19	904,113
Approved	66.44	67.96	69.35	69.81	1,984,308
N	506,809	596,174	860,520	924,918	2,888,421

APPLICANT INFORMATION

Applicant Sex

The sex variable was categorized as Female, Joint, Male, and Sex Not Specified. In 2018, 21.06% (106,742) of loans had a primary applicant identify as female, 32.82% (166,350) were applied by a primary applicant who identified as joint. 30.48% (154,494) identified as male, and 15.63% (79,223) had sex not specified.

Similarly, in 2019, 20.55% (122,529) of loans had a primary applicant identify as female, 32.78% (195,398) were joint applied by a primary applicant who identified as joint. 30.83% (183,800) identified as male, and 15.84% (94,447) had sex not specified.

In 2018, 19.39% (166,871) of loans had a primary applicant identify as female, 33.77% (290,595) were joint applied by a primary applicant who identified as joint. 30.73% (264,457) identified as male, and 16.11% (138,597) had sex not specified.

Finally, in 2021, 20.82% (192,557) of loans had a primary applicant identify as female, 32.00% (295,935) were joint applied by a primary applicant who identified as joint. 30.86% (285,394) identified as male, and 16.33% (151,032) had sex not specified.

The overall summary shows that across all years (2018 to 2021) shows that 20.38% (588,699) of loans had a primary applicant identify as female, 32.83% (948,278) were joint applied by a primary applicant who identified as joint. 30.75% (888,145) identified as male, and 16.04% (463,299) had sex not specified. (See Table 10 below)

Table 10 - Sex of the Applicant (North Carolina)

Sex	2018 %	2019 %	2020 %	2021 %	Total Period
Female	21.06	20.55	19.39	20.82	588,699
Male	30.48	30.83	30.73	30.86	888,145
Joint	32.82	32.78	33.77	32.00	948,278
Sex Not Specified	15.63	15.84	16.11	16.33	463,299
Total N	506,809	596,174	860,520	924,918	2,888,421

Co-Applicant Sex

Table 11 summarizes the distribution of the sex of the co-applicant which allows for the analysis of sex composition during the period in North Carolina and the 7 cities (Table 11.1-11.7) The category "Not applicable" refers to loan applications where having a co-applicant was not applicable or relevant. The category “No co-applicant” indicates loans that had only one primary applicant and this accounted for about half of the sample population under study. The largest sex composition for co-applicant were females and the accounted for about a quarter of the data collected.

Table 11 - Sex of the Co-Applicant (North Carolina)

Sex	2018 %	2019 %	2020 %	2021 %	Total Period
Male	8.59	8.92	9.35	9.62	266,108
Female	26.09	25.56	26.04	24.08	731,405
Information not provided by applicant	3.25	3.38	3.99	4.02	108,192
Not applicable	6.67	7.34	6.16	5.95	185,561
No co-applicant	55.36	54.78	54.44	56.30	1,596,422
Co-applicant selected both male and female	0.04	0.03	0.02	0.02	733
Total N	506,809	596,174	860,520	924,918	2,888,421

Applicant Ethnicity

The table below presents the distribution of both ethnicity and race among loan applicants in the state for the years 2018 to 2021. Unfortunately, about 21.9% and 0.04% (634,609 and 1114) of the population of the loan applicants failed to provide information on their ethnicity. Among the loan applicants for the period, 72.4% of the applicants were Not Hispanic, 4.2% were Hispanic and 1.36% were Joint.

For 2018, 4.06% of the applicants identified as Hispanic, 74.38% identified as non-Hispanic, 1.14% identified as joint indicating Hispanic ethnicity mixed with non-Hispanic ethnicity. 20.39% did not provide ethnicity, 0.03% associated with free form text only, implying that some borrowers provided ethnicity information in the form of text rather than selecting predefined categories. Similar patterns were observed for the years 2019 to 2021 (see Table 12 below).

Table 12 - Ethnicity of the Applicant (North Carolina)

Ethnicity	2018 %	2019 %	2020 %	2021 %	Total Period
Ethnicity Not Available	20.39	21.16	22.43	22.94	634,609
Free form Text only	0.03	0.04	0.04	0.04	1,114
Hispanic or Latino	4.06	4.27	4.03	4.51	122,479
Joint	1.14	1.23	1.39	1.44	38,308
Not Hispanic	74.38	73.30	72.12	71.07	2,091,911
Total N	506,809	596,174	860,520	924,918	2,888,421

Applicant Race

The race categories were specified as "White", "Black or African American," "American Indian or Alaska Native," "Asian," "Native Hawaiian or Other Pacific Islander," "2 or more minority races," "Joint ", "Free Form Text Only," and "Race Not Available". The analysis suggests that White borrowers were the most dominant racial group in the lending market during that period. However, it is important to note that the percentage of loan applicants identifying as White exhibited a gradual decline over the years, indicating a growing diversity among loan applicants.

Table 13 - Race of the Applicant (North Carolina)

Race	2018 %	2019 %	2020 %	2021 %	Total Period
2 or more minority races	0.18	0.16	0.16	0.19	4,977
American Indian or Alaska Native	0.60	0.59	0.53	0.58	16,455
Asian	3.17	3.19	4.02	4.39	110,344
Black or African American	11.57	11.35	9.86	10.98	312,796
Free Form Text Only	0.02	0.02	0.01	0.01	382
Joint	1.34	1.39	1.46	1.49	41,477
Native Hawaiian or Other Pacific Islander	0.17	0.16	0.14	0.14	4,214
Race Not Available	19.84	20.95	22.06	23.18	629,694
White	63.10	62.21	61.76	59.03	1,768,082
Total N	506,809	596,174	860,520	924,918	2,888,421

Table 14 - Race/Ethnicity Recode of the Applicant (North Carolina)

Race/Ethnicity Recode	2018 %	2019 %	2020 %	2021 %	Total Period
African American	11.31	11.08	9.58	10.65	304,270
American Indian	0.53	0.51	0.44	0.48	13,964
Asian	3.07	3.08	3.89	4.23	106,487
Hispanic	0.45	0.45	0.51	0.60	14,874
Multiracial Non-White	0.18	0.16	0.16	0.19	4,977
Multiracial White	1.34	1.39	1.46	1.49	41,477
Non-Hispanic White	63.10	62.21	61.76	59.03	1,768,082
Other/Free form Text	19.86	20.96	22.07	23.19	630,076
Pacific Islander	0.17	0.16	0.14	0.14	4,214
Total N	506,809	596,174	860,520	924,918	2,888,421

Race/Ethnicity Recoded

In order to facilitate meaningful analysis and ensure consistency in our dataset, we consolidated the race/ethnicity variables into a single categorical variable, to obtain a more streamlined and standardized representation of the racial and ethnic diversity in our dataset. This recoded variable will allow us to conduct more efficient and meaningful statistical analyses, ensuring that we can investigate any potential disparities and patterns in loan approvals across different racial and ethnic groups accurately. The summarized results can be found in Table, providing a clear understanding of the distribution of loan applications among the various race/ethnicity categories. The category "Hispanic" was treated as unique and retained separately, irrespective of the race combinations. Individuals identifying as "American Indian or Alaska Native" were combined into one category, while those identifying as "Asian" and "African American" were assigned to their respective groups.

Additionally, instances where individuals identified with multiple races involving Hispanic ethnicity, were categorized as "multiracial/Hispanic." On the other hand, cases of multiple races without any involvement of Hispanic ethnicity were labeled as "multiracial non-Hispanic." This approach ensured that the Hispanic ethnicity was consistently identified regardless of its combination with other races.

Among the different race/ethnicity groups, the largest proportion of loan applications came from Non-Hispanic Whites, accounting for around 63% of total applications in 2018 and gradually decreasing to 59.03% in 2021 as shown in Table

14 above. On the other hand, African American applicants comprised approximately 11% of loan applications in each year, while Asian applicants represented around 3-4% of the total.

The Hispanic category showed a slight increase in loan applications from 0.45% in 2018 to 0.60% in 2021. Similarly, the Multiracial White category displayed a steady increase in application percentages over the years, rising from 1.34% in 2018 to 1.49% in 2021.

Income

The income data presented in Table 15 shows the distribution of income for mortgage loan applicants in dollars ($) for the years 2018 to 2021. The dataset initially included negative values and non-numeric values “NA” for income, where NA indicated missing various which could be for various reasons such as non-response from applicants or data entry errors. We also recognized that negative income values were not meaningful or realistic in the context of personal income and decided to exclude these negative values from the dataset to ensure the accuracy and reliability of the results. For 2018 the average income of applicants was $100,160, this figure grew steadily over the next two years but fell in 2021 to $119,220.

Understanding the differences in mean income among racial groups can help identify potential disparities or patterns in income levels and access to credit. Table 16 below presents the mean income rounded to the whole number for different racial groups in dollars ($) for the period. We notice that in the year 2021 mean income for each race fell from that of 2020. This change could be attributed to various factors such as the economic impact of the COVID-19 pandemic, changes in employment and labor dynamics as a result of the pandemic, changes in sample composition and other socio-economic factors such as changes in policies and costs of living. We observe that Asians have the highest average income in the sample ($158,858 for the period) followed by applicants who selected joint as a race. Applicants who are American Indian or Alaska Native have the lowest mean income from 2018 to 2021 followed by Black or African American then Native Hawaiian or Other Pacific Islander.

Table 15 - Reported Family Income (North Carolina)

Amount	2018 ($)	2019 ($)	2020 ($)	2021 ($)	Total Period ($)
Mean	100,160	105,660	226,120	119,220	144,060
Median	73,000	77,000	87,000	85,000	82,000
Minimum	0	0	0	0	0
Maximum	136,107,000	85,000,000	894,000,000	582,937,000	894,000,000
N	454,366	522,313	728,337	795,937	2,500,953

Table 16 - Mean Family Income by Race (North Carolina)

Race	2018($)	2019($)	2020($)	2021($)	Total Period ($)
2 or more minority races	103,503	80,148	80,803	96,991	90,361
American Indian or Alaska Native	64,919	66,758	165,874	97,671	98,806
Asian	122,450	127,942	239,274	145,767	158,858
Black or African American	71,276	72,623	216,645	84,772	111,329
Free Form Text Only	84,441	92,911	76,879	92,471	86,676
Joint	119,293	123,114	234,279	139,731	154,104
Native Hawaiian or Pacific Islander	80,096	81,399	151,725	91,915	101,284
Race Not Available	108,625	111,830	215,541	128,870	141,216
White	102,701	109,081	230,075	120,511	140,592
Total N	454,366	522,313	728,337	795,937	2,500,953

Community Measures

The analysis of HMDA data in conjunction with community measures provides insights into the relationships between applicant income and various demographic indicators. Table 17 presents the mean values of community measures in North Carolina over the years 2018 to 2021.

The community measures, including census tract population, percent minority, tract to MSA median family income percentage (a measure of relative wealth), number of owner-occupied units, and the number of single-family units, were then analyzed alongside the applicant's income and the loan amount requested. This analysis aimed to explore the relationships between these community measures and the applicant's financial profile.

The population of North Carolina remained relatively stable during this period, with an average of 5,513 individuals. The percentage of minorities in the area ranged

from 29.63% to 30.31%, indicating a good consistent level of diversity. The tract to MSA income percent, which compares the tract's median family income to the metropolitan statistical area (MSA) median family income, showed some variation over the years but was generally increasing and averaged around 115.43%. The number of owner-occupied units also increased relatively over the years, with an average of 1,476 units.

Table 17 - Community Measures (North Carolina)

	2018	2019	2020	2021	Total Period
Population	5429	5470	5555	5546	5513
Percent Minority (%)	30.31	30.23	29.63	30.18	30.05
Tract to MSA Income Percent (%)	112.27	113.65	118.26	115.67	115.43
Owner Occupied Units	1450	1462	1493	1484	1476
Single Family (1-4)	0.998	0.998	0.998	0.998	0.998
Total N	506,809	596,174	860,520	924,918	2,888,421

Relationship between Population and Income

Table 18 shows the correlation coefficients assessed at a 5% significance level (2-tailed) between the community measures and the population per census block group in North Carolina. The results from the table indicate that there is a statistically significant linear relationship between the population and income. The correlation coefficient of -0.0014* suggests a small negative relationship, indicating that areas with a higher population tend to have slightly lower incomes. This relationship was found to be statistically significant ($p = 0.0257$).

Similarly, there is a statistically significant negative correlation between the percent minority and income, implying that areas with a higher percentage of minority residents tend to have lower incomes. The tract to MSA income percent shows a statistically significant positive correlation with income, suggesting that areas where the tract's median family income is higher relative to the MSA tend to have higher incomes.

On the other hand, no statistically significant correlation was found between owner-occupied units and income (correlation coefficient: 0.0007, $p = 0.2658$) and single-family units and income.

Likewise, for the loan amount, we observe a statistically significant negative correlation between population, percent minority, owner-occupied units, single family units, and loan amount. We however see a statistically significant positive relationship between tract to MSA income percent and loan amount.

These findings highlight the persistent links between geographical places and social indicators, which were key factors leading to the adoption of Fair Housing Laws and continued monitoring of HMDA data.

Table 18 - Correlation with Population Per Census Block Group (North Carolina)

	Correlation (p-value)	
Variable	Income	Loan Amount
Population	-0.0014* (0.0257)	-0.0059* (0.0000)
Percent Minority	-0.0008* (0.0000)	-0.0236* (0.0000)
Tract to MSA Income Percent	0.0156* (0.000)	0.0535* (0.0000)
Owner Occupied Units	0.0007 (0.2658)	-0.0026* (0.0000)
Single Family (1-4)	-0.0003 (0.5988)	-0.2802* (0.0000)

* Correlation is significant at the 0.05 level (2-tailed).

ANALYSIS OF LOAN APPROVAL VS. NON-APPROVAL

As previously mentioned, it is important to note that the loan action variable in the dataset was recoded to classify applications as "loan not approved" for specific reasons, including "application denied" (12.99%), "application withdrawn by applicant," "file closed for incompleteness," "preapproval request denied," or "preapproval request approved but not accepted." However, it should be noted that applications withdrawn by the applicant or closed for incompleteness, although not approved, are not considered as denied in the Loan Application Register (LAR). Consequently, no information regarding denial reasons for incompleteness or withdrawal was provided for these particular cases. This limitation results in missing information on denial reasons for a significant proportion of the dataset, as only 393,708 out of 904,113 applications were classified as denied and provided a basis for denial.

It is important to consider the reasons for denial to identify potential areas for improvement in the lending process. Table 19 provides an overview of the reasons for loan denial in North Carolina from 2018 to 2021. The most common reason for denial across all years was the debt-to-income ratio, which accounted for approximately 25.91% in 2018, 24.37% in 2019, 26.07% in 2020, and 26.73% in 2021. This indicates that a significant proportion of loan applicants were denied based on their level of debt relative to their income.

Another notable reason for denial was credit history, which accounted for a considerable percentage. In 2018, 30.98% of loan denials were attributed to credit history, which decreased to 28.27% in 2019, 27.19% in 2020, and 26.25% in 2021. This suggests that a significant number of applicants faced loan denials due to their past credit performance, such as a history of late payments or defaults.

Collateral or rather the lack thereof was also identified as a reason for loan denial, with percentages ranging from 15.02% in 2018 to 10.86% in 2021. This indicates that some applicants were unable to provide sufficient collateral to secure the loan, which potentially affected their chances of loan approval. Other reasons for denial, such as employment history, insufficient cash for down payment or closing costs, unverifiable information, and incomplete credit applications, also contributed to loan denials to varying but relatively low degrees.

The distribution of the reasons for loan denial for the seven cities under study are illustrated in the tables in the appendices. As we examine these tables, we observe a consistent pattern, mirroring the trends observed. For all seven cities, the prevalent reasons for denial over the years were Debt-to-income-ratio and Credit ratio accounting for over 20% in each instance.

These findings emphasize the significance of Debt-to-income ratio and Credit history as critical factors affecting loan approval outcomes in the studied towns. The consistent and substantial proportion of denials attributed to these factors suggests that they play a pivotal role in the decision-making process of loan approvals.

Table 19 - Reasons for Denial (North Carolina)

Loan Action	2018 %	2019 %	2020 %	2021 %	Total Period
Debt-to-income ratio	25.91	24.37	26.07	26.73	101,555
Employment history	1.21	1.04	1.52	1.27	4,979
Credit history	30.98	28.27	27.19	26.25	110,411
Collateral	15.02	13.08	11.34	10.86	49,050
Insufficient cash (down payment, closing costs)	2.45	2.10	2.00	1.67	8,008
Unverifiable information	5.28	4.95	5.63	6.28	21,888
Credit application incomplete	8.15	8.24	14.27	14.96	45,716
Mortgage insurance denied	0.06	0.02	0.02	0.03	119
Other	7.61	7.86	9.27	9.59	34,020
Not Applicable	3.34	10.07	2.69	2.36	17,962
Total N	87,552	96,880	102,648	106,628	393,708

Approvals by Type and Purpose of Loan

In this section, we focus on examining the approval rates of different loan types, shedding light on the viability and popularity of different financing options. For Conventional, FHA loans and VA loans about 63% to 70% of the loan approval rate indicating that over two thirds of applicants for these loan types were approved.

FSA/RHS loans, targeted towards rural borrowers and those with low-to-moderate incomes, displayed the highest approval rates among the loan types analyzed. Starting at 81.18% in 2018, the approval rates for FSA/RHS loans remained consistently high, reaching 82.41% in 2021 suggesting that these loans play a vital role in facilitating access to financing for rural communities and borrowers with specific income requirements in North Carolina.

Table 20 - Approval by Loan Type (North Carolina)

Loan Type	2018 %	2019 %	2020 %	2021 %	Total Period %
Conventional	66.34	67.78	68.57	69.60	68.37
FHA	63.75	66.26	69.29	68.02	67.02
VA	67.34	69.16	72.14	71.16	70.46
FSA/RHS	81.18	80.02	81.97	82.41	81.50

Approval by Applicant Characteristics

A bivariate analysis is adopted to investigate the influence of various applicant attributes on the loan approval process. We explore how individual characteristics, such as race, ethnicity, sex, and income, interact with loan approval rates.

Applicant Sex

The following tables present the approval rates by sex in North Carolina for the years 2018 to 2021. The data reveals interesting patterns in loan approvals based on sex of the applicant. Across all years, for North Carolina, we observe higher approval rates for male applicants compared to female applicants, with percentages ranging from 61.13% to 65.97% for males and 60.28% to 65.07% for females. Joint applicants consistently experienced the highest approval rates, ranging from 69.04% to 72.50%. Conversely, the not approved rates show a reverse trend, with female applicants consistently having slightly higher rates of loan denials compared to males and joint applicants.

A Pearson chi-square test was performed to test the association between sexes and loan approval outcomes. The chi-square test statistic is 163.6047, and the p-value associated with the test is 0.000 indicating that there is a strong statistically significant association between sex and loan approval outcomes. In other words, the proportion of loan approvals differs significantly between females and males.

Denial by Sex

Bivariate analysis of the denial reasons by sex helps shed light on the potential sex-specific factors contributing to loan denials in North Carolina. Table 22 helps us gain insights into the reasons for loan denials by sex in North Carolina for the years 2018 to 2021. It is evident that there are variations in the denial reasons between male and female applicants. Across all years, the most prominent reason for loan denial for both sexes are the "debt-to-income ratio," accounting for approximately 25% to 31.49% of denials for females and 24.46% to 31.49% for males. Additionally, "credit history" emerges as another significant factor leading to loan denials, with rates ranging from 25.74% to 33.21% for females and 26.48% to 30.68% for males.

On the other hand, the category "credit application incomplete" is more prevalent in denials for males (ranging from 6.71% to 16.20%) than females (ranging from 7.36% to 12.05%). Another interesting finding is that "collateral" plays a more significant role in female loan denials (ranging from 9.04% to 15.71%) compared to male denials (ranging from 9.38% to 12.38%).

Table 21 - Approval Rates by Sex (North Carolina)

Sex	2018 %		2019 %		2020 %		2021 %		Total Period	
	Approved	Not Approved	Approved	Not Approved	Approved	Not Approved	Approved	Not Approved	Approved	Not Approved
Female	60.28	39.72	62.17	37.83	64.29	35.71	65.07	34.93	373,095	215,604
Male	61.13	38.87	63.16	36.84	65.51	34.49	65.97	34.03	572,037	316,108
Joint	69.04	30.96	70.25	29.75	71.18	28.82	72.50	27.50	673,490	274,788

Table 22 - Denial Reasons by Sex (North Carolina)

Sex	2018		2019		2020		2021		Total	
	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female
Debt-to-income ratio	25.15	30.04	25.41	31.01	24.46	31.49	25.28	31.49	33,464	31,357
Employment history	1.32	1.12	1.32	1.04	1.63	1.59	1.46	1.28	1,923	1,280
Credit history	30.68	33.21	30.19	32.72	26.48	30.59	25.74	27.97	37,519	31,275
Collateral	15.71	12.38	14.45	11.14	11.79	9.04	11.46	9.38	17,640	10,504
Insufficient cash	2.71	2.41	2.44	2.12	2.09	2.02	1.81	1.60	2,984	2,035
Unverifiable information	5.16	5.12	5.33	5.27	5.86	5.58	6.30	5.95	7,604	5,562
Credit application incomplete	8.57	6.71	9.47	7.36	15.53	10.71	16.20	12.05	16,964	9,471
Mortgage insurance denied	0.09	0.05	0.02	0.02	0.03	0.01	0.02	0.03	53	27
Other	8.29	7.27	9.22	7.54	10.46	7.62	10.46	9.18	12,913	8,038
Not Applicable	2.32	1.68	2.14	1.78	1.68	1.34	1.26	1.07	2,430	1,459
Total	30,346	23,207	31,440	23,850	34,891	25,695	36,817	28,256	133,494	101,008

Approvals by Race/ Ethnicity

Table 23 provides a comprehensive overview of approval rates based on race in North Carolina over the four-year period. Among the racial categories, Asian applicants consistently exhibit the highest approval rates, ranging from 63.53% in 2018 to 67.83% in 2021 of their loan applications being approved. Conversely, African American applicants have the lowest approval rates, starting at 49.88% in 2018 and increasing slightly to 57.96% in 2021. Notably, White applicants consistently maintain approval rates above 67%, indicating a higher likelihood of loan approval compared to other racial groups.

Interestingly, the "Joint" category exhibited higher approval rates, surpassing even those of Non-Hispanic White applicants in some years, such as 69.14% in 2021. In contrast, applicants who identified with "2 or minority races" or "Native Hawaiian or Other Pacific Islander" experienced lower approval rates, consistently below 60% in most years.

To test the association between race/ethnicity and loan approval outcomes, a Pearson chi-square test was conducted. The chi-square test statistic was 4.2e+04 (42,000), and the p-value associated with the test is 0.000 indicating that the proportion of loan approvals differs significantly across different race/ethnicity categories.

Table 23 - Approval Rates by Race (North Carolina)

Race	2018 %		2019 %		2020 %		2021 %		Total	
	Approved	Not Approved	Approved	Not Approved	Approved	Not Approved	Approved	Not Approved	Approved	Not Approved
2 or minority races	39.96	60.04	46.17	53.83	49.23	50.77	52.47	47.53	2,396	2,581
American Indian or Alaska Native	43.71	56.29	46.22	53.78	48.32	51.68	49.72	50.28	7,811	8,644
Asian	63.53	36.47	64.76	35.24	64.73	35.27	67.83	32.17	72,499	37,845
Black or African American	49.88	50.12	53.16	46.84	56.41	43.59	57.96	42.04	171,994	140,802
Joint	65.04	34.96	66.21	33.79	67.44	32.56	69.14	30.86	27,941	13,536
Native Hawaiian or Other Pacific Islander	47.45	52.55	50.81	49.19	58.89	41.11	59.52	40.48	2,316	1,898
White	67.40	32.60	68.72	31.28	70.24	29.76	71.07	28.93	1,231,689	536,393

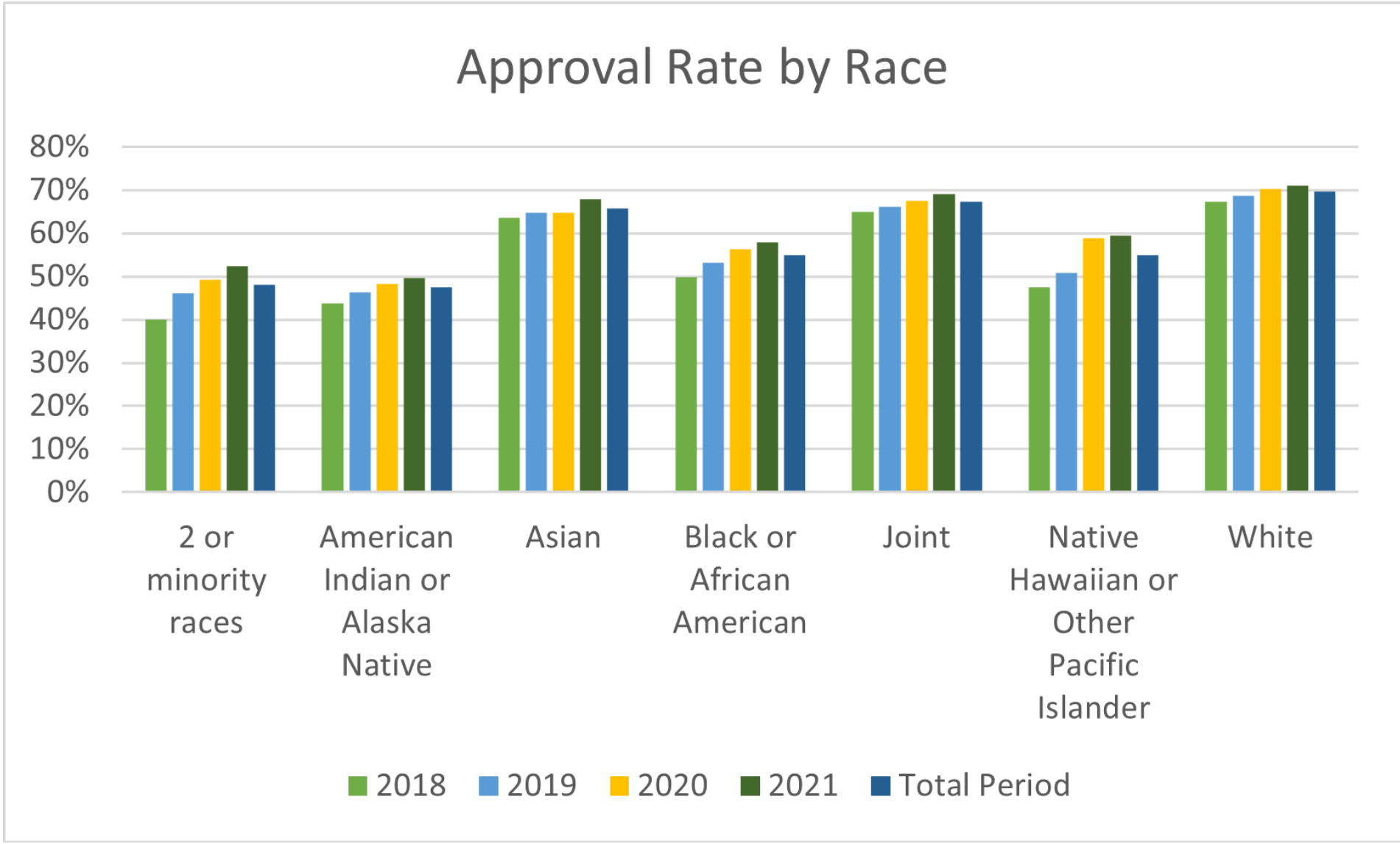


Figure 1 - Approval Rates by Race

Minority Status and Approval Rates

We conducted a comparative analysis denial reason between Non-Hispanic White applicants and Non-White applicants. Non-Hispanic White includes applicants who do not identify as minority but non-Hispanic, while Non-White includes Hispanic or Hispanic-identifying individuals. Analyzing the data, we observe interesting patterns in the denial reasons for both groups over the years. Notably, Non-White applicants consistently experience higher denial rates across various denial reasons compared to Non-Hispanic White applicants.

The most prevalent reason for loan denials in both groups is the "Debt-to-income ratio," with Non-White applicants consistently having higher denial rates than Non-Hispanic White applicants. This suggests that the Debt-to-income ratio is a significant factor affecting loan approvals for both groups, but it may be more pronounced for Non-White applicants.

Similarly, "Credit history" also stands out as another crucial factor contributing to loan denials for both groups. Again, Non-White applicants face higher denial rates in this category, indicating that Credit history plays a substantial role in the loan approval process, and it may disproportionately impact Non-White applicants.

In contrast, "Employment history" shows a more balanced distribution between the two groups, with minimal differences in denial rates. However, other denial reasons, such as "Collateral," "Insufficient cash," and "Unverifiable information," also exhibit varying degrees of discrepancy between the two groups, with Non-White applicants often experiencing higher denial rates.

The presence of the "Mortgage insurance denied" category exclusively in the Non-White group suggests that this denial reason may have unique implications for this subgroup. However, due to the relatively low count, further investigation may be required to draw definitive conclusions.

Table 25 below presents the distribution of denial reasons for various racial and ethnic groups based on their debt-to-income ratios. When comparing this table to the previously analyzed Table 24 above we observe some similarities. Notably, Asians stand out with 17.23% for "Credit History" denials, which is lower compared to other racial/ethnic categories. Despite this, across both tables, the most common denial reason for all racial/ethnic groups is Credit History and debt to income ratio accounting for a substantial proportion of denials.

Table 24 - Minority Status and Denial Reasons

Race	2018		2019		2020		2021		Total Period	
	Non-Hispanic White	Non-White	Non-Hispanic White	Non-White	Non-Hispanic White	Non-White	Non-Hispanic White	Non-White	Non-Hispanic White	Non-White
Debt-to-income ratio	23.57	26.39	19.11	27.20	22.49	27.57	22.39	28.60	1,183	24,705
Employment history	2.02	1.19	1.24	1.06	1.27	1.52	0.84	1.19	70	1,117
Credit history	35.71	37.12	38.39	38.50	33.67	34.67	37.61	32.43	1,958	31,933
Collateral	13.37	10.73	14.09	9.35	9.04	7.27	9.87	8.02	611	7,892
Insufficient cash	2.90	2.34	2.06	2.00	1.34	1.80	0.90	1.49	92	1,698
Unverifiable information	7.92	5.87	7.25	5.24	7.16	5.82	6.58	6.15	387	5,198
Credit application incomplete	5.45	6.67	7.66	6.61	13.59	11.57	11.68	11.50	539	8,290
Mortgage insurance denied	7.21	0.07	9.14	0.02	10.17	0.01	9.35	0.02	490	27
Other	0	7.73	0	8.28	0	8.54	0	9.52	0	7,693
Not Applicable	1.85	1.90	1.07	1.74	1.27	1.23	0.77	1.08	65	1,319
Total	1,137	21,071	1,214	21,350	1,494	22,570	1,550	24,881	5,395	89,872

Table 25 - Race and Ethnicity and Denial Reasons (North Carolina/2018 -2021)

(Race/Ethnicity)	Debt-to-income ratio	Employment history	Credit history	Collateral	Insufficient cash	Unverifiable information	Credit application incomplete	Mortgage insurance denied	Other	Not Applicable	Total
African American	26.97	1.10	38.47	8.65	1.74	5.57	7.95	0.03	8.36	1.17	69,634
American Indian	23.46	0.90	38.82	6.89	1.35	9.59	6.69	0.00	7.27	5.02	4,003
Asian	31.88	1.88	17.23	10.32	2.93	5.89	17.51	0.04	10.08	2.24	11,821
Hispanic	26.48	2.22	36.15	7.83	2.02	5.55	9.61	0.00	9.49	0.65	3,372
Multiracial Non-White	27.51	1.44	39.06	7.45	1.75	5.32	7.98	0.08	8.89	0.53	1,316
Multiracial White	21.93	1.30	36.29	11.33	1.71	7.17	9.99	0.00	9.08	1.20	5,395
Non-Hispanic White	26.18	1.28	27.51	14.19	2.21	5.58	11.47	0.03	8.90	2.66	222,869
Other/Free form Text	22.84	1.23	19.78	11.90	1.72	5.12	15.12	0.03	7.92	14.34	74,256
Pacific Islander	31.29	1.63	32.44	10.27	1.82	4.89	9.12	0.19	6.81	1.54	1,042
Total N	101,555	4,979	110,411	49,050	8,008	21,888	45,716	119	34,020	17,962	393,708

Approval by Income

For our analysis, it was important to investigate the relationship between loan approval and the loan-to-income ratio, that is, considering the impact of loan amounts requested on applications. As expected, we found that as income increased, the size of the loans also tended to increase, demonstrating a positive correlation ($r = 0.0087$, $p < 0.000$). To account for this effect, we calculated the loan-to-income ratio, which compares the loan amount requested to the reported income of the applicant.

The loan-to-income ratio was further standardized into Z-scores and categorized into five distinct groups based on specific ranges. These categories were as follows: loans with a ratio below the mean, loans with a ratio between mean income and +1 standard deviation above, loans with a ratio between +1 and +5 standard deviations above the mean, loans with a ratio between +5 and +20 standard deviations above the mean, and loans with a ratio more than +20 standard deviations above the mean.

In the dataset, there were 8,223 loan applications falling into the "+1 to +5 standard deviations above the mean" category, with 1,958 of them approved and 6,265 denied. For the "+5 to +20 standard deviations above the mean" category, there were 2,867 applications, with 340 approved and 2,527 denied. In the "Mean income to +1 standard deviation above the mean" category, there were 2,452,847 applications, with 1,643,119 approved and 809,728 denied. Lastly, for the "More than +20 standard deviations above the mean" category, there were 424,484 applications, with 338,891 approved and 85,593 denied.

Our statistical analysis shows that the association between loan approval and the loan-to-income ratio is statistically significant. The chi-square test statistic resulted in a large value of 4.0e+04 (40,000), and the corresponding p-value is 0.000 indicating that the proportion of loan approvals differs significantly across the different loan-to-income categories. This finding reiterates the importance of the loan-to-income ratio in the loan approval process, as it strongly influences the likelihood of loan approval. These results emphasize the need to consider the loan-to-income ratio as a crucial factor when evaluating loan applications and making well-informed lending decisions.

Table 26 - Loan to Income Ratio and Approval (North Carolina (2018-2021)

Loan to Income Ratio		Not Approved	Approved	Total
Below Mean Income	Count	-	-	-
	%	-	-	-
Mean income to +1 Standard Deviation	Count	6,265	1,958	8,223
	%	0.69	0.10	0.28
+1 to +5 Standard Dev	Count	2,527	340	2,867
	%	0.28	0.02	0.10
+5 to +20 Standard Dev	Count	809,728	1,643,119	2,452,847
	%	89.56	82.81	84.92
More than +20 Standard Deviations above the mean	Count	85,593	338,891	424,484
	%	9.47	17.08	14.70
Total	Count	904,113	1,984,308	2,888,421
	%	31.30	68.70	100

Approval by Community Measures

Exploring the means of community measures allows us to assess how various socioeconomic and demographic indicators within neighborhoods might contribute to differential loan approval rates. By comparing the means, we can identify any significant differences between the two groups, shedding light on the community-level factors that may influence lenders' decisions. Understanding these disparities can help us recognize potential patterns of discrimination or bias in the lending process.

For population, Tract to MSA income percent and owner-occupied units we notice that the mean for the approved applicants is slightly higher than the mean for denied applications. We performed a one-way ANOVA test which further confirmed the statistical significance of the differences in these community measures between the two groups (approved vs. denied loan applications). All ANOVA tests resulted in very low p-values ($p < 0.000$), indicating that there are significant differences in these community measures between the two groups. This reiterates the importance of controlling for community measures in our analysis.

Table 27 - Loan to Income Ratio and Approval

	Approval	N	Mean	Std. Deviation	Std Error Mean
Population	Approved	1984308	5578.181	2444.307	1.735207
	Not Approved	904113	5368.421	2520.359	2.650642
Percent Minority (%)	Approved	1984308	29.58935	20.72001	.0147091
	Not Approved	904113	31.05884	22.66074	.0238321
Tract to MSA Income Percent (%)	Approved	1984308	117.8438	42.56986	.0302202
	Not Approved	904113	110.1371	44.26226	.0465503
Owner Occupied Units	Approved	1984308	1498.403	702.0054	.4983516
	Not Approved	904113	1427.825	717.9131	.7550235
Single Family (1-4)	Approved	1984308	.9977317	.0475726	.0000338
	Not Approved	904113	.9989194	.032855	.0000346

ODDS OF APPROVAL: LOGISTIC REGRESSION ANALYSIS

The primary objective of this study is to examine the impact of race/ethnicity and/or sex of the applicant on mortgage application outcomes, specifically focusing on the presence of structural barriers to homeownership for minority groups for North Carolina over the years 2018 to 2021. To achieve this, logistic regression analysis is employed as the chosen methodological approach. Logistic regression is well-suited for modeling binary outcomes, such as loan approval or denial, which makes it ideal for investigating the relationship between the dependent variable, loan approval, and various independent variables.

The independent variables in this analysis encompass a range of factors that could potentially influence loan approval. These include indicators for loan characteristics such as type , purpose of loan and property type and indicators of applicant characteristics such as the sex and minority status of the individual .It also includes some continuous variables including the population, tract minority population, tract to MSA income percentage, number of dwellings lived in by the owner, and number of single family units provide insights into the neighborhood and market conditions surrounding the loan applications.

To account for potential confounding factors and better isolate the relationship between race/ethnicity, sex and loan approval, this study controls for various independent variables in the logistic regression analysis. By including these covariates, we seek to minimize the influence of alternative explanations and obtain a more accurate estimation of the impact of race/ethnicity on mortgage loan approvals.

Controlling these independent variables enables us to address the potential confounding effects arising from loan-specific characteristics and community measures. This approach allows us to focus on evaluating the specific influence of race/ethnicity and sex on loan approval. This comprehensive approach enhances the robustness of the analysis and provides a clearer understanding of whether any structural barriers or disparities associated with race/ethnicity and sex play a role in affecting access to mortgage credit.

By employing logistic regression, we can estimate the probabilities of loan approval based on the independent variables and examine the significance of their effects. The coefficients obtained from the regression output show how each independent

variable relates to the log-odds of loan approval. Positive coefficients indicate a higher chance of approval, while negative coefficients suggest a lower likelihood. We determine the statistical significance of each variable using the associated p-values, where smaller values indicate a stronger case against the null hypothesis that race/ethnicity and sex do not affect loan approval. Any observed differences in approval rates among racial or ethnic groups and sex are unlikely due to random chance or unrelated factors.

Initially, we encountered challenges with the logistic regression model convergence. After thorough investigation, we identified that the issue primarily stemmed from the presence of high and right-skewed income, loan amount, and loan-to-income ratio values in our HMDA dataset. We observed a significant number of loan amounts exceeding $1 million (16,123 loans) and high-income values surpassing $100 million (387,468 applicants). Additionally, there were a substantial number of loan-to-income ratios that were exceptionally high, exceeding +20 standard deviations above the mean, with a count of 431,965 instances.

To address this, we performed a logarithmic transformation on income, loan amount, and loan-to-income ratios to achieve model convergence. This transformation effectively reduced the impact of extreme values and skewed distributions, making the variables more suitable for logistic regression analysis. The log transformation allows us to better capture the underlying relationships between these variables and the likelihood of loan approval. By mitigating the influence of outliers and extreme values, our logistic regression model now provides reliable and interpretable results, enabling us to draw meaningful conclusions about the determinants of loan approval outcomes in our study.

By conducting this pre-processing step of logarithmic transformation, we enhance the robustness of our analysis and ensure that our findings are more representative of the broader population. This approach also makes our results less susceptible to undue influence from extreme observations, including outliers allowing us to derive more accurate insights into the factors influencing loan approval outcomes.

Table 28 - Logistic Regression Tables (North Carolina)

	Model 1		Model 2		Model 3	
	β	Standard Error	β	Standard Error	β	Standard Error
Constant	0.0633	.1924429	0.0720	.1921843	2.265***	.1905448
Loan Characteristics						
Home Improvement/ Refinancing	-0.535***	.0029063	-0.534***	.0029077	-0.559***	.0028863
Not Owner Occupied	-0.189***	.0067009	-0.194***	.006733	-0.128***	.0066827
Manufactured	-1.300***	.0064936	-1.302***	.0065001	-1.526***	.0063842
Non-Conventional loans	-0.228***	.0038322	-0.223***	.0038561	-0.258***	.003833
Loan Amount (Log)	0.221***	.0019961	0.219***	.0020031	-	-
Community Measures						
Population	0.0000173***	1.52e-06	0.0000163***	1.52e-06	0.0000260***	1.51e-06
Percent Minority	0.00109***	.0000854	0.00122***	.0000857	0.000178**	.000085
Tract to MSA Income	0.000992***	.0000391	0.000962***	.0000392	0.00299***	.0000375
Tract Owner Occupied Units	0.0000367***	5.26e-06	0.0000405***	5.27e-06	-5.74e-09	5.23e-06
Single Family Units (1-4 units)	-1.383***	.191982	-1.380***	.1917156	-1.555***	.1904272
Applicant Characteristics						
Female	0.0365***	.00346	0.0439***	.0034797	-0.0924***	.0033797
Minority	-0.434***	.0037662	-	-	-	-
Hispanic	-	-	-0.577***	.0179994	-0.597***	.0178473
Asian	-	-	-0.386***	.007102	-0.305***	.0070464
African American	-	-	-0.481***	.0046318	-0.505***	.0045999
Other races	-	-	-0.271***	.0094661	-0.236***	.0093821
Income (Log)	0.244***	.0023489	0.243***	.0023493	-	-
Loan to Income Ratio (Log)	-	-	-	-	0.0362***	.0017632
Pseudo R squared	5.36%		5.38%		4.27%	
Log Likelihood	-1483987.5		-1483715.4		-1500986	
Model Chi-square	168033.73***		168577.93***		134036.76***	
Degree of Freedom1	13		16		15	
N	2,465,455		2,465,455		2,465,455	

* p<0.10, ** p<0.05, *** p<0.01

coefficient to percentage formula: $(e^{\beta} - 1) * 100\%$.

Model 1: Approval Odds for Minority vs. Non-Hispanic White Populations

In the first model, we examined all the independent variables to establish a baseline understanding of their significance in Table 28 below. With the exception of tract owner-occupied units, all other independent variables demonstrated statistical significance at various levels.

With the loan characteristics, when controlling for other factors, we observed a 41.43% decrease in the probability of loan approval for home improvements or refinancing compared to loans for home purchase and other purposes. Additionally, the probability of loan approval for manufactured homes was 72.74% lower than that for single-family housing. 17.22% Moreover, loans categorized as FHA, VA, or FHS had a 20.39% higher likelihood of approval compared to conventional loans.

Moving on to the neighborhood characteristics, we discovered that most community variables included in the analysis did not exhibit the strong impact that was initially expected. While all variables demonstrated statistical significance and had measurable effects on loan approval probabilities, their overall impact was not as influential as hypothesized. For each one unit the increase in population led to a 0.0173% rise in the probability of loan approval. Similarly, each non-white individual within the census tract increased the probability of loan approval by 0.109%. Furthermore, as the percentage of the tract's income relative to the MSA median income increased by one percent (above the mean of 115.43), the likelihood of loan approval increased by 0.099%. The impact for these variables was not as powerful as anticipated. Conversely, loans for single-family units experienced a 74.92% decrease in the probability of approval compared to multifamily dwellings.

In the individual characteristics of the primary applicant, we observed that both minority status and the sex of the applicant were statistically significant predictors of loan approval probability. Minority applicants had a 35.2% lower likelihood of loan approval compared to non-Hispanic White applicants. Additionally, contrary to our expectations, non-female applicants had a 3.72% lower probability of loan approval compared to female applicants.

The extremely low p-value of 0.0000 indicates that the overall model is statistically significant, suggesting that at least one of the predictors is significantly associated with the outcome. These findings suggest the presence of structural barriers in the mortgage market, leading to disparities in loan approval rates based on race/ethnicity and sex in the first model. However, the pseudo-R squared of 0.0536, which is a measure of how much of the variation in the outcome variable is explained by the model, indicates that there are likely other factors not captured by the model that contribute to the loan approval process. While 5.36% may seem relatively low, it is common for logistic regression models to have lower R-squared values compared to linear regression models due to the nature of the binary outcome variable.

Overall, the analysis suggests that the model has a statistically significant relationship with the outcome variable, and the model explains a modest proportion of the variance in the outcome.

Model 2: Approval Odds by Distinct Race/Ethnic Groups vs. Non-Hispanic White Populations

In the second model, we aim to delve deeper into the loan approval process by incorporating additional insights related to the race/ethnicity of the primary applicant. Alongside loan characteristics, community characteristics, and applicant characteristics, this expanded model provides a more comprehensive understanding of the factors influencing mortgage loan outcomes.

By including race/ethnicity as a variable, we seek to identify any potential disparities or unique patterns associated with different and/or specific racial or ethnic groups' loan approval rates. This analysis allows us to assess whether race/ethnicity plays a significant role in determining the likelihood of loan approval, beyond the effects of other variables.

With the loan characteristics, when controlling for other factors, we observe that there is a decrease in the probability that loans for home improvements or refinancing (41.37%), or not to be owner occupied (17.6%), or for manufactured houses (72.8%), or FHA, VA, or FHS loans (20%), would be approved.

Similarly, as seen in the first model we notice that most community variables included in the analysis did not exhibit the strong impact that was initially expected. While all the variables in this block demonstrated statistical significance and had measurable effects on loan approval probabilities, their overall impact was not as

influential as hypothesized. For each one unit increase in population led to a 0.002% rise in the probability of loan approval. Similarly, each non-white individual within the census tract increased the probability of loan approval by 0.122%. Furthermore, as the percentage of the tract's income relative to the MSA median income increased by one percent (above the mean of 115.43), the likelihood of loan approval increased by 0.096%.

Finally in the third block, we observed that the sex of the applicant and all the race/ethnicity variables were statistically significant predictors of loan approval probability. Non-female applicants had a 4.5% lower probability of loan approval compared to female applicants. Hispanic applicants were less likely to get their loan approved (43.84%) than non-Hispanic applicants when all other factors are controlled for. Likewise Asian applicants and African American Applicants were less likely to have their mortgage loan applications approved (32% and 38.2%) than their non-Asian and non-African American counterparts.

These findings suggest the presence of structural barriers in the mortgage market, leading to disparities in loan approval rates based on race/ethnicity and sex in the first model. However, the pseudo-R squared of 0.0538 indicating that the model explains approximately 5.38% of the total variance in the outcome variable. While 5.38% may seem relatively low, it is common for logistic regression models to have lower R-squared values compared to linear regression models due to the nature of the binary outcome variable. The extremely low p-value indicated that the model is statistically significant which suggests that the variables considered are valuable predictors of loan approval, and the model provides important insights into how different factors influence the likelihood of loan approval when controlling for other variables.

Model 3: Approval Odds by Distinct Race/Ethnic Groups vs. Non-Hispanic White Populations Accounting for LTV

To enhance the understanding of the mortgage loan approval process we incorporated the logarithmically transformed loan-to-income ratio as a key financial capacity indicator. This transformation allows us to account for the non-linear relationship between loan-to-income ratio and loan approval probabilities, providing a more accurate representation of how borrowers' financial capacity influences the likelihood of loan approval. The added variable loan to income ratio clearly shows that the odds of approval increase significantly (3.68%) for every one-unit increase in the log loan-to-income ratio. For example, a house that is three

times the reported family income would be 3.68% more likely to be approved than one that is two times the family income. This means that borrowers with higher log loan-to-income ratios are more likely to have their mortgage loan applications approved compared to borrowers with lower log loan-to-income ratios when all other factors in the model are held constant.

However, it's crucial to acknowledge that the overall pseudo-R squared value of 0.0427 indicates that the model explains only about 4.27% of the total variance in the loan approval outcome. While this might seem relatively low, it's a common occurrence in logistic regression models due to the binary nature of the outcome variable. It implies that there are other unobserved factors contributing to the loan approval process beyond what's included in the model.

Additionally, our analysis underscored the significance of race/ethnicity and sex as key predictors of loan approval probabilities. Non-female applicants had an 8.8% lower likelihood of loan approval compared to female applicants, and notable disparities in approval rates were evident among different racial and ethnic groups. Hispanic applicants, for instance, were less likely to have their loans approved (44.95%) than non-Hispanic applicants, with similar trends observed for Asian and African American applicants. These findings strongly suggest the presence of structural barriers and potential disparities in the mortgage market based on race/ethnicity and sex .

In conclusion, our logistic regression analysis offered valuable insights into the factors influencing mortgage loan approval rates. The inclusion of the log loan-to-income ratio provided a clearer understanding of borrowers' financial capacity impact, while the significant role of race/ethnicity and sex shed light on potential disparities. By recognizing these patterns, we can work towards promoting a fair and equitable mortgage lending process for all applicants. Nonetheless, it's important to recognize the model's limitations, as it only explains a portion of the loan approval process, leaving room for other unobserved factors to contribute to the outcomes.

Regressions by City

Logistic regressions were then conducted, adapting all model adaptations to focus on the seven previously mentioned cities. The results in Tables 29.1-29.3 reaffirm the significance of specific factors in influencing loan approval probabilities. Across all cities, loan characteristics play a critical role in determining approval rates. Home Improvement/Refinancing and Manufactured loans consistently have a lower likelihood of approval compared to other loan types. Non-Conventional loans also show reduced odds of approval in most cities.

Community measures have varying effects on loan approval probabilities across different cities. While population and tract to MSA income ratios show some impact, the influence of the percentage of minority individuals differs between cities. Moreover, single-family units generally experience a decrease in the probability of approval across most cities.

Applicant characteristics demonstrate interesting trends as well. Minority status consistently leads to a lower likelihood of loan approval in all cities. However, the effect of sex on approval rates varies between cities, with some cities showing higher approval odds for females, while others exhibit the opposite.

The pseudo-R-squared values indicate that the model explains a modest proportion of the variance in loan approval outcomes. This suggests that there may be other factors not considered in the analysis that also contribute to loan approval disparities.

The findings suggest the presence of potential structural barriers in the mortgage market, leading to disparities in loan approval rates based on various factors, including loan type, minority status, and location.

Table 29 - Logistic Regression Model 1 by City

	Raleigh (β)	Durham (β)	Greensboro (β)	Winston-Salem (β)	Charlotte (β)	Asheville (β)	Wilmington (β)
Constant	-0.631	-0.983	1.166	1.590	0.446	-1.007	-1.458***
Loan Characteristics							
Home Improvement/ Refinancing	-0.444***	-0.442***	-0.655***	-0.648***	-0.484***	-0.300***	-0.644***
Not Owner Occupied	-0.0707***	-0.127***	-0.247***	-0.394***	-0.115***	-0.250***	-0.463***
Manufactured	-1.148***	-0.849***	-1.478***	-1.327***	-1.121***	-1.074***	-0.660***
Non-Conventional loans	-0.393***	-0.460***	-0.259***	-0.265***	-0.339***	-0.326***	-0.00972
Loan Amount (Log)	0.261***	0.253***	0.210***	0.181***	0.291***	0.219***	0.262***
Community Measures							
Population	-0.00000357	0.0000161***	-0.00000566	0.0000166*	-0.00000272	-0.0000268***	0.0000636
Percent Minority	-0.00320***	0.000174	0.00237***	-0.00129***	-0.00200***	0.00585***	-0.000826
Tract to MSA Income	-0.00144***	0.000621***	-0.000207	0.000437**	-0.00113***	0.000204	0.000388
Tract Owner Occupied Units	0.0000163	-0.0000347**	0.0000250	-0.0000155	0.0000215*	0.000206***	-0.0000117
Single Family Units (1-4 units)	-0.321	-0.426	-2.171**	-2.658**	-1.538**	-0.384	0
Applicant Characteristics							
Female	0.0645***	0.0466***	0.0914***	0.0645***	0.0588***	0.0221	-0.0832*
Minority	-0.414***	-0.382***	-0.359***	-0.389***	-0.339***	-0.355***	-0.622***
Income (Log)	0.247***	0.274***	0.297***	0.312***	0.209***	0.234***	0.197***
Pseudo R squared	3.58%	4.25%	5.44%	4.88%	3.80%	3.96%	5.16%
Log Likelihood	-251266.79	-88881.596	-87479.065	-79875.405	-373520.11	-73467.259	-10735.871
Model Chi-square	18656.77***	7888.26***	10071.08***	8204.01***	29495.96***	6063.11***	1167.55***
Degree of Freedom	13	13	13	13	13	13	13
N	430,938	152,767	143,791	132,512	620,017	120772	17647

* p<0.10, ** p<0.05, *** p<0.01

Table 30 - Logistic Regression Model 2 by City

	Raleigh (β)	Durham (β)	Greensboro (β)	Winston-Salem (β)	Charlotte (β)	Asheville (β)	Wilmington (β)
Constant	-0.630	-0.936	1.165	1.597	0.464	-1.015	-1.446***
Loan Characteristics							
Home Improvement/ Refinancing	-0.444***	-0.436***	-0.648***	-0.648***	-0.483***	-0.300***	-0.644***
Not Owner Occupied	-0.0690***	-0.140***	-0.246***	-0.396***	-0.121***	-0.249***	-0.463***
Manufactured	-1.148***	-0.848***	-1.479***	-1.328***	-1.121***	-1.075***	-0.650***
Non-Conventional loans	-0.391***	-0.447***	-0.260***	-0.264***	-0.332***	-0.326***	-0.0120
Loan Amount (Log)	0.261***	0.248***	0.210***	0.180***	0.289***	0.220***	0.261***
Community Measures							
Population	-0.00000370	0.0000129**	-0.00000576	0.0000160*	-0.00000411	-0.0000269***	0.0000572
Percent Minority	-0.00305***	0.000499	-0.00230***	-0.00120***	-0.00186***	0.00600***	0.0000293
Tract to MSA Income	-0.00143***	0.000545***	-0.000194	0.000434**	-0.00114***	0.000223	0.000409
Tract Owner Occupied Units	0.0000174	-0.0000235	0.0000254	-0.0000140	0.0000257**	0.000207***	0.00000146
Single Family Units (1-4 units)	-0.322	-0.441	-2.170**	-2.661**	-1.548**	-0.382	0
Applicant Characteristics							
Female	0.0720***	0.0603***	0.0940***	0.0658***	0.0662***	0.0259*	-0.0702
Hispanic	-0.497***	-0.708***	-0.565***	-0.777***	-0.428***	-0.786***	-0.657*
Asian	-0.421***	-0.300***	-0.381***	-0.354***	-0.294***	-0.500***	-0.618***
African American	-0.449***	-0.454***	-0.363***	-0.395***	-0.379***	-0.426***	-0.806***
Other races	-0.200***	-0.100***	-0.215***	-0.304***	-0.195***	-0.176***	-0.270**
Income (Log)	0.245***	0.270***	0.296***	0.311***	0.208***	0.233***	0.193***
Pseudo R squared	3.60%	4.31%	5.45%	4.90%	3.81%	3.98%	5.23%
Log Likelihood	-251216.75	-88826.343	-87469.243	-79863.233	-373474.05	-73451.887	-10727.539
Model Chi-square	18756.86***	7998.77***	10090.73***	8228.36***	29588.07***	6093.85***	1184.22***
Degree of Freedom	16	16	16	16	16	16	16
N	430,938	152,767	143,791	132,512	620,017	120772	17647

* p<0.10, ** p<0.05, *** p<0.01

Table 31 - Logistic Regression Model 3 by City

	Raleigh (β)	Durham (β)	Greensboro (β)	Winston-Salem (β)	Charlotte (β)	Asheville (β)	Wilmington (β)
Constant	1.921***	1.936**	3.384***	3.713***	3.018***	1.200	0.355***
Loan Characteristics							
Home Improvement/ Refinancing	-0.478***	-0.476***	-0.664***	-0.652***	-0.511***	-0.363***	-0.687***
Not Owner Occupied	0.0132	-0.0688***	-0.150***	-0.306***	-0.0376***	-0.204***	-0.428***
Manufactured	-1.431***	-1.135***	-1.669***	-1.502***	-1.361***	-1.320***	-0.842***
Non-Conventional loans	-0.400***	-0.475***	-0.257***	-0.272***	-0.356***	-0.356***	0.0429
Community Measures							
Population	0.0000116***	0.0000322***	0.00000949	0.0000295***	-0.00000312	-0.0000190**	0.000157**
Percent Minority	-0.00446***	-0.00143***	-0.00340***	-0.00276***	-0.00210***	0.00596***	-0.00218
Tract to MSA Income	0.000965***	0.00243***	0.00239***	0.00255***	0.00126***	0.00215***	0.00644***
Tract Owner Occupied Units	-0.0000488***	-0.000116***	-0.00000503	-0.0000636**	0.00000942	0.000162***	-0.000421*
Single Family Units (1-4 units)	-0.583	-0.861	-2.404**	-2.756***	-1.959***	-0.549	0
Applicant Characteristics							
Female	-0.0679***	-0.0781***	-0.0414***	-0.0779***	-0.0701***	-0.111***	-0.198***
Hispanic	-0.518***	-0.733***	-0.599***	-0.806***	-0.458***	-0.784***	-0.649*
Asian	-0.374***	-0.237***	-0.375***	-0.297***	-0.232***	-0.430***	-0.510***
African American	-0.468***	-0.508***	-0.363***	-0.392***	-0.401***	-0.482***	-0.827***
Other races	-0.151***	-0.0307	-0.180***	-0.239***	-0.160***	-0.160***	-0.233**
Loan to Income Ratio (Log)	0.0634***	0.0361***	0.0191***	0.00517	0.0943***	0.0208***	0.0504**
Pseudo R squared	2.53%	3.02%	4.24%	3.70%	2.56%	2.93%	4.34%
Log Likelihood	-253998.6	-90026.708	-88596.583	-80870.524	-378330.34	-74260.234	-10828.203
Model Chi-square	13193.15***	5598.04***	7836.05***	6213.77***	19875.50***	4477.16***	982.89***
Degree of Freedom	15	15	15	15	15	15	15
N	430,938	152,767	143,791	132,512	620,017	120772	17647

* p<0.10, ** p<0.05, *** p<0.01

SUMMARY & CONCLUSIONS

In this report, we have examined housing mortgage loan application data for the state of North Carolina via logistic regression models across three primary blocks of variables: loan characteristics, community characteristics, and applicant characteristics.

In the first model, without including race/ethnicity as a variable, loan characteristics such as home improvements or refinancing, owner occupancy, and being manufactured homes, and FHA, VA, or FHS loans all had statistically significant impacts on the likelihood of loan approval, with a decrease in probability of approval. Meanwhile, community variables (e.g., population, non-white population percentage, tract's income relative to the MSA median income) exhibited less of an impact on loan approval probability than was initially expected. The sex of the applicant was also a statistically significant predictor of loan approval probability, with non-female applicants having a lower probability of loan approval.

In the expanded model that included race/ethnicity as a variable, the trends seen in the first model largely persisted, with the additional finding that Hispanic, Asian, and African American applicants were statistically less likely to have their mortgage loan applications approved than their non-Hispanic, non-Asian, and non-African American counterparts. This suggested the presence of structural barriers in the mortgage market, leading to disparities in loan approval rates based on race/ethnicity and sex.

In the third model, a log-transformed loan-to-income ratio was introduced as an indicator of a borrower's financial capacity, which showed that the odds of loan approval increase for higher loan-to-income ratios. The results also reinforced the findings about the impact of race/ethnicity and sex on loan approval probabilities.

Finally, the models were adapted to focus on seven specific cities, reaffirming the significance of certain factors in influencing loan approval probabilities. Across all cities, certain loan characteristics were consistently linked with lower approval rates. Community measures and applicant characteristics, including minority status, had varied effects across cities.

The pseudo-R-squared values across all models were modest, indicating that while the models had some predictive power, there are likely other unobserved factors

influencing loan approval rates. The analyses suggest potential structural barriers in the mortgage market that contribute to disparities in loan approval rates based on loan type, minority status, and location.

The table provided on the next page offers insights into the probability of mortgage approval for various demographic groups, taking White males as the reference category. Analyzing this data through the lens of a logistic regression model, we find that differences do exist when we compare approval rates across different racial and gender groups, holding all other variables constant.

For instance, White females exhibit a 4.49% higher probability of mortgage approval as compared to White males. On the other hand, the likelihood for Asian males and females is decreased by 32.02% and 28.97% respectively. Even steeper declines are evident among Hispanic males and females, with their probabilities lower by 43.84% and 41.32%, respectively. Black males and females also face decreased probabilities, lower by 38.18% and 35.41% respectively, compared to White males. Finally, males and females from other ethnic backgrounds have probabilities reduced by 23.74% and 20.32% respectively, when compared with White males.

These findings, though primarily statistical associations, allude to potential disparities in mortgage approval rates across different racial and gender groups. It is crucial to underline that while these percentages indicate variations in approval probabilities, they do not automatically signal causal relationships.

These results, however, do not stand in isolation. They are echoed by numerous peer-reviewed studies which have highlighted racial and gender disparities in mortgage approvals. For instance, a study by Ross and Yinger (2002) found that Black and Hispanic applicants were more likely to be denied credit than comparable White applicants. Similarly, a report by Turner et al. (2002) highlighted that women, regardless of race, faced significant barriers in securing mortgages compared to their male counterparts.

Taken together, these findings call for further inquiry into factors contributing to such disparities and underscore the need for strategies to promote equitable access to credit for all demographic groups.

Table 32 - Model 2 Comparison of Odds Ratio as Percent Likelihood of Approval

	Constant (β)	Female (β)	Hispanic (β)	Asian (β)	African American (β)	Other (β)	Sum	Exp (β)	Probability of Approval Relative to White Men	Percent Difference from White Men
White M ale	0.0720						0.072	1.075		
White Female	0.0720	0.0439					0.116	1.123	1.0449	4.49%
Asian Male	0.0720			-0.386			-0.314	0.731	0.680	-32.02%
Asian Female	0.0720	0.0439		-0.386			-0.270	0.763	0.710	-28.97%
Hispanic Male	0.0720		-0.577				-0.505	0.604	0.562	-43.84%
Hispanic Female	0.0720	0.0439	-0.577				-0.461	0.631	0.587	-41.32%
Black Male	0.0720				-0.481		-0.409	0.664	0.618	-38.18%
Black Female	0.0720	0.0439			-0.481		-0.365	0.694	0.646	-35.41%
Other Male	0.0720					-0.271	-0.199	0.820	0.763	-23.74%
Other Female	0.0720	0.0439				-0.271	-0.155	0.856	0.797	-20.32%

Rounded to the 3rd decimal place.

RECOMMENDATIONS

The findings reveal disparities in mortgage loan approval rates based on factors such as race/ethnicity, gender, and certain loan and community characteristics. The implications are significant as they point towards potential structural barriers within the mortgage lending process. Such disparities could exacerbate social inequalities and hinder economic mobility for certain groups, particularly minorities.

Here are several ways in which cities and communities could promote more equity in residential mortgage lending:

- *Fair Housing and Lending Practices*: Cities can do more to enforce fair housing and lending laws to prevent discrimination based on race, ethnicity, gender, and other protected classes. This could include more rigorous audits of lending institutions and consequences for those found to be engaging in discriminatory practices.

- *Homeownership Education and Counseling:* Establish programs that offer education and counseling to prospective minority homeowners. This could help individuals better understand the mortgage process, improve their credit scores, and increase their chances of loan approval.

- *Financial Assistance Programs:* Implement financial assistance programs that help minority applicants with down payments and closing costs. This could increase the accessibility of homeownership for minority groups.

- *Community Reinvestment Act (CRA):* Local governments could advocate for stricter enforcement of the CRA, which encourages banks to meet the credit needs of all residents, including those in low- and moderate-income neighborhoods.

- *Affordable Housing Initiatives:* Cities can encourage the development of affordable housing units. This can be achieved through zoning modifications, public-private partnerships, and subsidies or tax incentives for developers willing to build affordable housing.

- *Loan Products for Lower-Income Borrowers:* Encourage the development and promotion of loan products designed specifically for lower-income borrowers, such as those with lower down payment requirements or more flexible credit score criteria.

- *Partnerships with Community Development Financial Institutions (CDFIs):* Cities can partner with CDFIs, which often have a mission to serve low-income or disadvantaged communities. They can offer a range of services, from personal finance counseling to small business lending, and could be enlisted to help increase homeownership in underserved communities.

To increase minority homeownership, existing programs such as the HUD's Homeownership Voucher Program could also be expanded. Other initiatives could include programs specifically targeting first-time minority homebuyers, or "rent-to-own" programs that allow renters to build equity over time. It's also important to address systemic issues related to income and wealth inequality, as these are key factors influencing homeownership rates.

Down payment assistance programs in particular, like those of the North Carolina Housing Finance Agency, are pivotal in bolstering minority homeownership as they address numerous challenges in the home buying process. These programs alleviate the most prevalent barrier to homeownership for many minority families: the substantial upfront cost of a down payment. By supplying grants or low-interest loans for this purpose, the financial hurdles are significantly reduced. In addition, these initiatives contribute to overall housing affordability, allowing minority families to consider homes in neighborhoods with greater resources and opportunities. This not only encourages upward mobility but also aids in wealth accumulation. When the need for a large down payment is reduced or removed, the loan-to-value ratio drops correspondingly. As a result, lenders perceive lower risk, potentially leading to improved mortgage loan approval rates and more favorable interest rates. Furthermore, down payment assistance enhances financial stability post-home purchase. Given that the initial financial burden is reduced, families are better equipped to manage unexpected expenses or financial emergencies, decreasing the likelihood of future foreclosure.

Ultimately, achieving more equitable outcomes in residential mortgage lending requires comprehensive and coordinated efforts across all levels of government, as well as active partnerships with the private sector and community organizations.

APPENDIX A – LOAN TYPE

Table 33 - Type of Loan (Raleigh)

Loan Type	2018 %	2019 %	2020 %	2021 %	Total Period
Conventional	80.10	79.20	81.80	85.03	404,170
FHA	9.63	9.69	7.21	6.19	38,247
VA	8.14	9.61	9.45	7.74	42,941
FSA/RHS	2.13	1.50	1.54	1.04	7,233
Total N	79,621	97,742	156,253	158,975	492,591

Table 34 - Type of Loan (Durham)

Loan Type	2018 %	2019 %	2020 %	2021 %	Total Period
Conventional	84.51	82.51	84.45	87.39	145,932
FHA	9.18	9.39	7.19	5.90	12,975
VA	5.86	7.51	7.70	6.22	11,822
FSA/RHS	0.45	0.59	0.66	0.49	7,233
Total N	27,540	36,383	52,246	55,518	171,687

Table 35 - Type of Loan (Greensboro/High Point)

Loan Type	2018 %	2019 %	2020 %	2021 %	Total Period
Conventional	73.69	73.43	75.44	78.53	125,442
FHA	16.11	15.60	13.11	11.81	22,818
VA	7.99	9.52	9.67	8.44	14,825
FSA/RHS	2.22	1.44	1.78	1.21	2,664
Total N	30,149	34,796	48,177	52,627	165,749

Table 36 - Type of Loan (Winston-Salem)

Loan Type	2018 %	2019 %	2020 %	2021 %	Total Period
Conventional	72.56	72.86	75.08	78.44	114,678
FHA	16.18	15.54	13.00	11.48	20,771
VA	8.84	10.04	10.26	8.98	14,539
FSA/RHS	2.41	1.57	1.66	1.11	2,443
Total N	28,237	31,643	43,439	49,112	152,431

Table 37 - Type of Loan (Charlotte)

Loan Type	2018 %	2019 %	2020 %	2021 %	Total Period
Conventional	77.62	77.42	80.23	80.23	575,056
FHA	14.18	13.53	10.70	10.70	82,760
VA	7.19	8.40	8.37	8.37	55,743
FSA/RHS	1.02	0.65	0.70	0.70	4,871
Total N	124,025	150,457	217,780	226,168	718,430

Table 38 - Type of Loan (Asheville)

Loan Type	2018 %	2019 %	2020 %	2021 %	Total Period
Conventional	82.44	81.75	84.17	86.53	113,199
FHA	8.61	8.35	6.13	5.69	9,312
VA	6.66	8.14	8.20	6.83	10,063
FSA/RHS	2.29	1.75	1.50	0.94	2,055
Total N	24,852	28,227	39,513	42,037	134,629

Table 39 - Type of Loan (Wilmington)

Loan Type	2018 %	2019 %	2020 %	2021 %	Total Period
Conventional	62.37	54.92	58.51	63.06	13,742
FHA	14.04	13.60	9.55	9.77	2,531
VA	21.73	30.07	30.60	25.99	6,355
FSA/RHS	1.85	1.40	1.34	1.19	314
Total N	3,083	4,559	7,323	7,977	22,942

APPENDIX B – PROPERTY TYPE

Table 40 - Property Type (Raleigh)

Property Type	2018 %	2019 %	2020 %	2021 %	Total Period
Multifamily: Manufactured	0.01	0.01	0.00	0.00	29
Multifamily: Site-Built	0.16	0.13	0.09	0.09	542
Single Family (1-4 Units): Manufactured	2.05	1.71	1.33	1.42	7,649
Single Family (1-4 Units): Site-Built	93.80	94.23	95.52	95.22	484,371
Total N	79,621	97,742	156,253	158,975	492,591

Table 41 - Property Type (Durham)

Property Type	2018 %	2019 %	2020 %	2021 %	Total Period
Multifamily: Manufactured	0.01	0.01	0.01	0.01	18
Multifamily: Site-Built	0.34	0.24	0.15	0.16	353
Single Family (1-4 Units): Manufactured	2.43	2.79	2.38	2.47	4,299
Single Family (1-4 Units): Site-Built	97.21	96.95	97.47	97.35	167,017
Total N	27,540	36,383	52,246	55,518	171,687

Table 42 - Property Type (Greensboro/High Point)

Property Type	2018 %	2019 %	2020 %	2021 %	Total Period
Multifamily: Manufactured	0.02	0.01	0.00	0.00	13
Multifamily: Site-Built	0.37	0.32	0.32	0.25	509
Single Family (1-4 Units): Manufactured	3.91	3.89	3.98	4.26	6,692
Single Family (1-4 Units): Site-Built	95.70	95.78	95.70	95.48	158,535
Total N	30,149	34,796	48,177	52,627	165,749

Table 43 - Property Type (Winston-Salem)

Property Type	2018 %	2019 %	2020 %	2021 %	Total Period
Multifamily: Manufactured	0.02	0.03	0.03	0.04	47
Multifamily: Site-Built	0.33	0.31	0.28	0.32	471
Single Family (1-4 Units): Manufactured	3.45	3.35	3.66	3.93	5,558
Single Family (1-4 Units): Site-Built	96.20	96.31	96.03	95.70	146,355
Total N	28,237	31,643	43,439	49,112	152,431

Table 44 - Property Type (Charlotte)

Property Type	2018 %	2019 %	2020 %	2021 %	Total Period
Multifamily: Manufactured	0.00	0.01	0.00	0.01	41
Multifamily: Site-Built	0.20	0.16	0.12	0.11	1,017
Single Family (1-4 Units): Manufactured	1.95	1.85	1.70	2.01	13,459
Single Family (1-4 Units): Site-Built	97.85	97.97	98.17	97.87	703,913
Total N	124,025	150,457	217,780	226,168	718,430

Table 45 - Property Type (Asheville)

Property Type	2018 %	2019 %	2020 %	2021 %	Total Period
Multifamily: Manufactured	0.06	0.03	0.02	0.02	39
Multifamily: Site-Built	0.28	0.18	0.16	0.10	222
Single Family (1-4 Units): Manufactured	6.92	6.20	6.57	7.82	9,353
Single Family (1-4 Units): Site-Built	92.74	93.59	93.26	92.06	125,015
Total N	24,852	28,227	39,513	42,037	134,629

Table 46 - Property Type (Wilmington)

Property Type	2018 %	2019 %	2020 %	2021 %	Total Period
Multifamily: Manufactured	0.03	0.00	0.01	0.00	2
Multifamily: Site-Built	0.06	0.04	0.01	0.01	6
Single Family (1-4 Units): Manufactured	9.08	6.95	5.11	5.63	1,420
Single Family (1-4 Units): Site-Built	90.82	93.00	94.87	94.36	21,514
Total N	3,083	4,559	7,323	7,977	22,942

APPENDIX C– PURPOSE

Table 47 - Purpose of the loan (Raleigh)

Purpose of Loan	2018 %	2019 %	2020 %	2021 %	Total Period
Home purchase	56.03	47.12	32.35	32.19	192,380
Home improvement	7.65	6.63	3.86	4.68	26,042
Other purpose	9.40	7.62	3.86	4.32	27,844
Not applicable	1.35	0.94	1.08	0.27	4,118
Refinancing	12.96	24.45	47.70	40.89	173,750
Cash-out Refinancing	12.60	13.24	11.16	17.64	68,457
Total N	79,621	97,742	156,253	158,975	492,591

Table 48 - Purpose of the loan (Durham)

Purpose of Loan	2018 %	2019 %	2020 %	2021 %	Total Period
Home purchase	52.38	46.13	33.52	32.32	66,665
Home improvement	8.43	7.61	4.38	4.66	9,963
Other purpose	9.66	8.28	4.27	4.45	10,376
Not applicable	1.19	0.98	1.13	0.35	1,468
Refinancing	15.41	23.35	45.54	41.37	59,499
Cash-out Refinancing	12.93	13.65	11.16	16.86	23,716
Total N	27,540	36,383	52,246	55,518	171,687

Table 49 -- Purpose of the loan (Greensboro/High Point)

Purpose of Loan	2018 %	2019 %	2020 %	2021 %	Total Period
Home purchase	50.42	48.01	36.78	35.95	68,543
Home improvement	7.45	7.23	4.78	5.00	9,696
Other purpose	9.92	8.61	5.44	4.82	11,144
Not applicable	2.12	1.87	2.48	0.62	2,812
Refinancing	17.06	21.33	39.71	36.59	50,953
Cash-out Refinancing	13.02	12.96	10.80	17.03	22,601
Total N	30,149	34,796	48,177	52,627	165,749

Table 50 - Purpose of the loan (Winston-Salem)

Purpose of Loan	2018 %	2019 %	2020 %	2021 %	Total Period
Home purchase	50.86	47.96	37.44	35.26	63,119
Home improvement	7.73	7.01	4.83	5.19	9,049
Other purpose	9.74	8.71	5.20	5.14	10,290
Not applicable	1.92	1.43	2.09	0.60	2,197
Refinancing	17.00	22.33	39.73	37.28	47,437
Cash-out Refinancing	12.75	12.57	10.70	16.53	20,339
Total N	28,237	31,643	43,439	49,112	152,431

Table 51 - Purpose of the loan (Charlotte)

Purpose of Loan	2018 %	2019 %	2020 %	2021 %	Total Period
Home purchase	52.74	46.25	33.43	32.44	281,164
Home improvement	7.48	6.72	3.97	4.52	38,254
Other purpose	8.65	7.35	3.98	4.14	39,833
Not applicable	1.42	1.28	1.21	0.31	7,039
Refinancing	13.91	22.65	43.59	37.83	231,816
Cash-out Refinancing	15.81	15.75	13.82	20.75	120,324
Total N	124,025	150,457	217,780	226,168	718,430

Table 52 - Purpose of the loan (Asheville)

Purpose of Loan	2018 %	2019 %	2020 %	2021 %	Total Period
Home purchase	48.94	44.50	34.97	36.11	53,719
Home improvement	9.29	8.07	5.33	5.43	8,976
Other purpose	9.48	8.01	4.66	4.23	8,239
Not applicable	0.68	0.62	0.71	0.34	762
Refinancing	14.92	21.72	38.78	34.51	39,674
Cash-out Refinancing	16.69	17.08	15.54	19.38	23,259
Total N	24,852	28,227	39,513	42,037	134,629

Table 53 - Purpose of the loan (Wilmington)

Purpose of Loan	2018 %	2019 %	2020 %	2021 %	Total Period
Home purchase	56.76	48.37	40.54	40.44	10,150
Home improvement	5.84	4.61	2.72	3.07	834
Other purpose	6.55	5.04	2.65	2.61	834
Not applicable	1.52	1.14	1.15	0.29	206
Refinancing	14.92	24.96	42.56	36.97	7,664
Cash-out Refinancing	14.40	15.88	10.38	16.62	3,254
Total N	3,083	4,559	7,323	7,977	22,942

APPENDIX D – OCCUPANCY

Table 54 - Owner Occupancy (Raleigh)

Occupancy Type	2018 %	2019 %	2020 %	2021 %	Total Period
Principal residence	92.82	93.66	94.43	93.33	459,736
Secondary residence	1.45	1.33	1.10	1.24	6,130
Investment Property	5.73	5.01	4.47	5.43	26,725
Total N	79,621	97,742	156,253	158,975	492,591

Table 55 - Owner Occupancy (Durham)

Occupancy Type	2018 %	2019 %	2020 %	2021 %	Total Period
Principal residence	90.89	91.88	92.36	90.91	157,185
Secondary residence	2.11	1.93	1.91	1.82	3,287
Investment Property	7.00	6.19	5.74	7.27	11,215
Total N	27,540	36,383	52,246	55,518	171,687

Table 56 - Owner Occupancy (Greensboro/High Point)

Occupancy Type	2018 %	2019 %	2020 %	2021 %	Total Period
Principal residence	93.37	93.76	94.45	93.57	155,524
Secondary residence	1.48	1.25	1.05	1.18	2,012
Investment Property	5.15	4.98	4.50	5.24	8,213
Total N	30,149	34,796	48,177	52,627	165,749

Table 57 - Owner Occupancy (Winston-Salem)

Occupancy Type	2018 %	2019 %	2020 %	2021 %	Total Period
Principal residence	94.03	94.51	94.79	94.14	143,871
Secondary residence	1.70	1.51	1.60	1.53	2,407
Investment Property	4.27	3.98	3.61	4.32	6,153
Total N	28,237	31,643	43,439	49,112	152,431

Table 58 - Owner Occupancy (Charlotte)

Occupancy Type	2018 %	2019 %	2020 %	2021 %	Total Period
Principal residence	92.40	93.16	93.97	92.44	668,490
Secondary residence	1.81	1.58	1.43	1.46	11,041
Investment Property	5.79	5.26	4.60	6.10	38,899
Total N	124,025	150,457	217,780	226,168	718,430

Table 59 - Owner Occupancy (Asheville)

Occupancy Type	2018 %	2019 %	2020 %	2021 %	Total Period
Principal residence	88.43	88.69	89.10	87.85	119,145
Secondary residence	6.00	5.85	6.16	6.32	8,234
Investment Property	5.57	5.46	4.74	5.83	7,250
Total N	24,852	28,227	39,513	42,037	134,629

Table 60 - Owner Occupancy (Wilmington)

Occupancy Type	2018 %	2019 %	2020 %	2021 %	Total Period
Principal residence	77.39	82.15	78.07	78.21	18,087
Secondary residence	15.93	12.48	17.74	16.46	3,672
Investment Property	6.68	5.37	4.19	5.33	1,183
Total N	3,083	4,559	7,323	7,977	22,942

APPENDIX E – LOAN AMOUNT

Table 61 - Loan Amount (Raleigh)

Loan Amount	2018 ($)	2019 ($)	2020 ($)	2021 ($)	Total Period
Mean	232,096.1	262,993.2	268,103.7	282,265.9	265,840.1
Median	195,000	225,000	245,000	255,000	235,000
Minimum	5,000	5,000	5,000	5,000	5,000
Maximum	150,000,000	900,000,000	51,400,000	114,000,000	900,000,000
N	79,621	97,742	156,253	158,975	492,591

Table 62 - Loan Amount (Durham)

Loan Amount	2018 ($)	2019 ($)	2020 ($)	2021 ($)	Total Period
Mean	243,144.2	255,959.8	264,618	282,493.1	265,118.8
Median	185,000	205,000	225,000	235,000	215,000
Minimum	5,000	5,000	5,000	5,000	5,000
Maximum	75,200,000	365,000,000	75,000,000	136,000,000	365,000,000
N	27,540	36,383	52,246	55,518	171,687

Table 63 - Loan Amount (Greensboro/High Point)

Loan Amount	2018 ($)	2019 ($)	2020 ($)	2021 ($)	Total Period
Mean	158,420.7	171,396.7	196,676.5	194,859.6	183,834
Median	125,000	135,000	155,000	165,000	145,000
Minimum	5,000	5,000	5,000	5,000	5,000
Maximum	34,600,000	50,000,000	46,300,000	50,000,000	50,000,000
N	30,149	34,796	48,177	52,627	165,749

Table 64 - Loan Amount (Winston-Salem)

Loan Amount	2018 ($)	2019 ($)	2020 ($)	2021 ($)	Total Period
Mean	153,149.2	170,203.4	188,954.7	196,091.6	180,728.8
Median	125,000	135,000	155,000	165,000	145,000
Minimum	5,000	5,000	5,000	5,000	5,000
Maximum	29,300,000	47,200,000	45,600,000	54,900,000	54,900,000
N	28,237	31,643	43,439	49,112	152,431

Table 65 - Loan Amount (Charlotte)

Loan Amount	2018 ($)	2019 ($)	2020 ($)	2021 ($)	Total Period
Mean	227,224.8	246,730.6	265,352.2	280,285.8	259,571.5
Median	175,000	195,000	225,000	235,000	215000
Minimum	5,000	5,000	5,000	5,000	5,000
Maximum	360,000,000	460,000,000	71,400,000	366,000,000	460,000,000
N	124,025	150,457	217,780	226,168	718,430

Table 66 - Loan Amount (Asheville)

Loan Amount	2018 ($)	2019 ($)	2020 ($)	2021 ($)	Total Period
Mean	207,640.4	227,857.2	248,651.7	263,772.3	241,442.6
Median	175,000	195,000	215,000	225,000	205,000
Minimum	5,000	5,000	5,000	5,000	5,000
Maximum	48,700,000	46,800,000	95,600,000	56,300,000	95,600,000
N	24,852	28,227	39,513	42,037	134,629

Table 67 - Loan Amount (Wilmington)

Loan Amount	2018 ($)	2019 ($)	2020 ($)	2021 ($)	Total Period
Mean	248,347.4	268,272.6	298,648.8	326,955.6	295,695.2
Median	225,000	255,000	275,000	285,000	265,000
Minimum	5,000	5,000	5,000	5,000	5,000
Maximum	3,525,000	7,675,000	5,435,000	9,105,000	9,105,000
N	3,083	4,559	7,323	7,977	22,942

APPENDIX F – PRE-APPROVAL

Table 68 - Pre-approval (Raleigh)

Pre-approval Status	2018 %	2019 %	2020 %	2021 %	Total Period
Pre-approval requested	1.40	1.57	0.88	0.92	5,484
Pre-approval not requested	98.60	98.43	99.12	99.08	487,107
N	79,621	97,742	156,253	158,975	492,591

Table 69 - Pre-approval (Durham)

Pre-approval Status	2018 %	2019 %	2020 %	2021 %	Total Period
Pre-approval requested	1.27	1.37	0.79	0.82	1,714
Pre-approval not requested	98.73	98.63	99.21	99.18	169,973
N	27,540	36,383	52,246	55,518	171,687

Table 70 - Pre-approval (Greensboro/High Point)

Pre-approval Status	2018 %	2019 %	2020 %	2021 %	Total Period
Pre-approval requested	1.17	1.70	1.12	1.34	2,186
Pre-approval not requested	98.83	98.30	98.88	98.66	163,563
N	30,149	34,796	48,177	52,627	165,749

Table 71 - Pre-approval (Winston-Salem)

Pre-approval Status	2018 %	2019 %	2020 %	2021 %	Total Period
Pre-approval requested	1.38	1.68	1.18	1.10	1,974
Pre-approval not requested	98.62	98.32	98.82	98.90	150,457
N	28,237	31,643	43,439	49,112	152,431

Table 72 - Pre-approval (Charlotte)

Pre-approval Status	2018 %	2019 %	2020 %	2021 %	Total Period
Pre-approval requested	1.59	1.44	0.82	0.93	8,029
Pre-approval not requested	98.41	98.56	99.18	99.07	710,401
N	124,025	150,457	217,780	226,168	718,430

Table 73 - Pre-approval (Asheville)

Pre-approval Status	2018 %	2019 %	2020 %	2021 %	Total Period
Pre-approval requested	1.01	1.17	0.72	0.73	1,169
Pre-approval not requested	98.99	98.83	99.28	99.27	133,460
N	24,852	28,227	39,513	42,037	134,629

Table 74 - Pre-approval (Wilmington)

Pre-approval Status	2018 %	2019 %	2020 %	2021 %	Total Period
Pre-approval requested	2.34	1.10	0.41	0.43	186
Pre-approval not requested	97.66	98.90	99.59	99.57	22,756
N	3,083	4,559	7,323	7,977	22,942

APPENDIX G – CROSSTABULATION OF APPROVAL VS. PRE-APPROVAL REQUEST

Table 75 - Crosstabulation of Approval Vs. Pre-Approval Request (Raleigh)

Pre-approval Status	2018 %	2019 %	2020 %	2021 %	Total Period
Pre-approval requested					5,484
Loan Approved	1,098 (98.74)	1,500 (97.91)	1,354 (98.19)	1,354 (92.68)	5,306 (96.75)
Not Approved	14 (1.26)	32 (2.09)	25 (1.81)	107 (7.32)	178 (3.25)
Pre-approval not requested					487,107
Loan Approved	55,416 (70.59)	68,769 (71.48)	111,713 (72.13)	114,553 (72.73)	350,451 (71.95)
Not Approved	23,093 (29.41)	27,441 (28.52)	43,161 (27.87)	42,961 (27.27)	136,656 (28.05)
N	79,621	97,742	156,253	158,975	492,591

Table 76 - Crosstabulation of Approval Vs. Pre-Approval Request (Durham)

Pre-approval Status	2018 %	2019 %	2020 %	2021 %	Total Period
Pre-approval requested					1,714
Loan Approved	345 (98.29)	489 (98.19)	400 (97.32)	415 (91.41)	1649 (96.21)
Not Approved	6 (1.71)	9 (1.81)	11 (2.68)	39 (8.59)	65 (3.79)
Pre-approval not requested					487,107
Loan Approved	18,976 (69.79)	25,056 (69.82)	37,174 (71.72)	40,055 (72.74)	121,261 (71.34)
Not Approved	8,213 (30.21)	10,829 (30.18)	14,661 (28.28)	15,009 (27.26)	48,712 (28.66)
N	27,540	36,383	52,246	55,518	171,687

Table 77 - Crosstabulation of Approval Vs. Pre-Approval Request (Greensboro/High Point)

Pre-approval Status	2018 %	2019 %	2020 %	2021 %	Total Period
Pre-approval requested					2,186
Loan Approved	339 (96.03)	573 (97.12)	524 (97.04)	627 (89.19)	2,063 (94.37)
Not Approved	14 (3.97)	17 (2.88)	16 (2.96)	76 (10.81)	123 (5.63)
Pre-approval not requested					163,563
Loan Approved	19,664 (66.00)	22,986 (67.20)	32,201 (67.60)	35,223 (67.84)	110,074 (67.30)
Not Approved	10,132 (34.00)	11,220 (32.80)	5,436 (32.40)	16,701 (32.16)	53,489 (32.70)
N	30,149	34,796	48,177	52,627	165,749

Table 78 - Crosstabulation of Approval Vs. Pre-Approval Request (Winston-Salem)

Pre-approval Status	2018 %	2019 %	2020 %	2021 %	Total Period
Pre-approval requested					1,974
Loan Approved	386 (98.97)	522 (98.12)	495 (96.30)	474 (88.10)	1,877 (95.09)
Not Approved	4 (1.03)	10 (1.88)	19 (3.70)	64 (11.90)	97 (4.91)
Pre-approval not requested					163,563
Loan Approved	18,610 (66.83)	21,379 (68.72)	29,495 (68.71)	33,399 (68.76)	102,883 (68.38)
Not Approved	9,237 (33.17)	9,732 (31.28)	13,430 (31.29)	15,175 (31.24)	47,574 (31.62)
N	28,237	31,643	43,439	49,112	152,431

Table 79 - Crosstabulation of Approval Vs. Pre-Approval Request (Charlotte)

Pre-approval Status	2018 %	2019 %	2020 %	2021 %	Total Period
Pre-approval requested					8,029
Loan Approved	1,951 (98.63)	2,109 (97.55)	1,728 (96.75)	1,903 (90.49)	7,691 (95.79)
Not Approved	27 (1.37)	53 (2.45)	58 (3.25)	200 (9.51)	338 (4.21)
Pre-approval not requested					710,401
Loan Approved	81,368 (66.67)	101,372 (68.36)	151,555 (70.17)	158,286 (70.64)	492,581 (69.34)
Not Approved	40,679 (33.33)	46,923 (31.64)	64,439 (29.83)	65,779 (29.36)	217,820 (30.66)
N	124,025	150,457	217,780	226,168	718,430

Table 80 - Crosstabulation of Approval Vs. Pre-Approval Request (Asheville)

Pre-approval Status	2018 %	2019 %	2020 %	2021 %	Total Period
Pre-approval requested					1,169
Loan Approved	240 (95.62)	314 (95.44)	268 (94.37)	267 (87.54)	1,089 (93.16)
Not Approved	11 (4.38)	15 (4.56)	16 (5.63)	38 (12.46)	80 (6.84)
Pre-approval not requested					133,460
Loan Approved	16,440 (66.83)	19,007 (68.13)	26,926 (68.64)	28,646 (68.64)	91,019 (68.20)
Not Approved	8,161 (33.17)	8,891 (31.87)	12,303 (31.36)	13,086 (31.36)	42,441 (31.80)
N	24,852	28,227	39,513	42,037	134,629

Table 81 - Crosstabulation of Approval Vs. Pre-Approval Request (Wilmington)

Pre-approval Status	2018 %	2019 %	2020 %	2021 %	Total Period
Pre-approval requested					186
Loan Approved	70 (97.22)	45 (90.00)	27 (90.00)	30 (88.24)	172 (92.47)
Not Approved	2 (2.78)	5 (10.00)	3 (10.00)	4 (11.76)	14 (7.53)
Pre-approval not requested					22,756
Loan Approved	1,988 (66.02)	3,119 (69.17)	5,171 (70.90)	5,631 (70.89)	15,909 (69.91)
Not Approved	1,023 (33.98)	1,390 (30.83)	2,122 (29.10)	2,312 (29.11)	6,847 (30.09)
N	3,083	4,559	7,323	7,977	22,942

APPENDIX H – LOAN ACTION

Table 82 - Loan Action (Raleigh)

Loan Action	2018 %	2019 %	2020 %	2021 %	Total Period
Loan originated	56.66	58.07	59.86	61.13	292,590
Application approved but not accepted	1.80	1.75	1.97	2.03	9,449
Application denied	12.96	11.63	9.26	8.96	50,393
Application withdrawn by applicant	12.53	12.47	13.25	12.17	62,208
File closed for incompleteness	3.51	3.98	5.12	5.90	24,055
Purchased loan	12.52	12.07	10.53	9.75	53,718
Preapproval request denied	0.01	0.01	0.01	0.02	62
Preapproval request approved but not accepted	0.01	0.02	0.01	0.05	116
Total N	79,621	97,742	156,253	158,975	492,591

Table 83 - Loan Action (Durham)

Loan Action	2018 %	2019 %	2020 %	2021 %	Total Period
Loan originated	56.96	57.21	59.29	60.87	101,274
Application approved but not accepted	2.04	1.92	1.91	2.05	3,397
Application denied	13.84	13.23	9.91	9.22	18,915
Application withdrawn by applicant	12.48	12.54	12.90	11.77	21,272
File closed for incompleteness	3.50	4.00	5.25	6.05	8,525
Purchased loan	11.16	11.08	10.72	9.97	18,239
Preapproval request denied	0.00	0.01	0.01	0.01	16
Preapproval request approved but not accepted	0.02	0.02	0.01	0.06	49
Total N	27,540	36,383	52,246	55,518	171,687

Table 84 - Loan Action (Greensboro/High Point)

Loan Action	2018 %	2019 %	2020 %	2021 %	Total Period
Loan originated	53.55	55.01	57.11	59.40	94,057
Application approved but not accepted	2.21	2.10	2.32	2.39	3,773
Application denied	18.26	15.87	13.17	13.00	24,208
Application withdrawn by applicant	11.69	12.06	13.36	11.90	20,422
File closed for incompleteness	3.67	4.32	5.51	6.83	8,859
Purchased loan	10.58	10.60	8.50	6.33	14,307
Preapproval request denied	0.04	0.04	0.03	0.05	69
Preapproval request approved but not accepted	0.00	0.01	0.00	0.09	54
Total N	30,149	34,796	48,177	52,627	165,749

Table 85 - Loan Action (Winston-Salem)

Loan Action	2018 %	2019 %	2020 %	2021 %	Total Period
Loan originated	55.61	57.54	59.61	60.89	89,710
Application approved but not accepted	1.88	1.83	2.12	2.34	3,185
Application denied	16.89	15.02	12.56	12.45	21,094
Application withdrawn by applicant	12.41	12.06	13.26	12.21	19,080
File closed for incompleteness	3.41	3.67	5.10	6.23	7,400
Purchased loan	9.77	9.84	7.31	5.74	11,865
Preapproval request denied	0.01	0.02	0.04	0.03	41
Preapproval request approved but not accepted	0.00	0.01	0.00	0.10	56
Total N	28,237	31,643	43,439	49,112	152,431

Table 86 - Loan Action (Charlotte)

Loan Action	2018 %	2019 %	2020 %	2021 %	Total Period
Loan originated	53.66	55.36	59.18	60.03	414,506
Application approved but not accepted	2.07	2.27	2.53	2.52	17,180
Application denied	16.04	14.07	10.85	10.73	88,961
Application withdrawn by applicant	12.88	12.91	13.32	12.21	92,031
File closed for incompleteness	3.88	4.20	5.42	6.14	36,828
Purchased loan	11.44	11.15	8.67	8.28	68,586
Preapproval request denied	0.00	0.01	0.01	0.02	102
Preapproval request approved but not accepted	0.02	0.02	0.01	0.07	236
Total N	124,025	150,457	217,780	226,168	718,430

Table 87 - Loan Action (Asheville)

Loan Action	2018 %	2019 %	2020 %	2021 %	Total Period
Loan originated	54.99	55.58	58.32	58.49	76,987
Application approved but not accepted	1.95	2.30	2.35	2.42	3,076
Application denied	16.63	14.41	12.19	11.81	17,981
Application withdrawn by applicant	12.35	12.76	13.04	12.42	17,044
File closed for incompleteness	3.85	4.33	5.91	6.89	7,416
Purchased loan	10.18	10.57	8.15	7.87	12,045
Preapproval request denied	0.02	0.02	0.03	0.01	26
Preapproval request approved but not accepted	0.02	0.03	0.01	0.08	54
Total N	24,852	28,227	39,513	42,037	134,629

Table 88 - Loan Action (Wilmington)

Loan Action	2018 %	2019 %	2020 %	2021 %	Total Period
Loan originated	48.26	51.22	56.04	55.71	12,371
Application approved but not accepted	1.43	1.43	1.87	2.29	429
Application denied	15.83	13.31	10.46	9.95	2,655
Application withdrawn by applicant	13.82	12.79	13.67	13.05	3,051
File closed for incompleteness	3.54	4.39	4.85	5.98	1,141
Purchased loan	17.06	16.76	13.07	12.96	3,281
Preapproval request denied	0.00	0.02	0.04	0.01	5
Preapproval request approved but not accepted	0.06	0.09	0.00	0.04	9
Total N	3,083	4,559	7,323	7,977	22,942

APPENDIX I – LOAN APPROVAL

Table 89 - Loan Approval (Raleigh)

Loan Action	2018 %	2019 %	2020 %	2021 %	Total Period
Not approved (Denied, Withdrawn, or Incomplete)	29.02	28.11	27.64	27.09	136,834
Approved	70.98	71.89	72.36	72.91	355,757
Total N	79,621	97,742	156,253	158,975	492,591

Table 90 - Loan Approval (Durham)

Loan Action	2018 %	2019 %	2020 %	2021 %	Total Period
Not approved (Denied, Withdrawn, or Incomplete)	29.84	29.79	28.08	27.10	48,777
Approved	70.16	70.21	71.92	72.90	122,910
Total N	27,540	36,383	52,246	55,518	171,687

Table 91 - Loan Approval (Greensboro/High Point)

Loan Action	2018 %	2019 %	2020 %	2021 %	Total Period
Not approved (Denied, Withdrawn, or Incomplete)	33.65	32.29	32.07	31.88	53,612
Approved	66.35	67.71	67.93	68.12	112,137
Total N	30,149	34,796	48,177	52,627	165,749

Table 92 - Loan Approval (Winston-Salem)

Loan Action	2018 %	2019 %	2020 %	2021 %	Total Period
Not approved (Denied, Withdrawn, or Incomplete)	32.73	30.79	30.96	31.03	47,671
Approved	67.27	69.21	69.04	68.97	104,760
Total N	28,237	31,643	43,439	49,112	152,431

Table 93 - Loan Approval (Charlotte)

Loan Action	2018 %	2019 %	2020 %	2021 %	Total Period
Not approved (Denied, Withdrawn, or Incomplete)	32.82	31.22	29.62	29.17	218,158
Approved	67.18	68.78	70.38	70.83	500,272
Total N	124,025	150,457	217,780	226,168	718,430

Table 94 - Loan Approval (Asheville)

Loan Action	2018 %	2019 %	2020 %	2021 %	Total Period
Not approved (Denied, Withdrawn, or Incomplete)	32.88	31.55	31.18	31.22	42,521
Approved	67.12	68.45	68.82	68.78	92,108
Total N	24,852	28,227	39,513	42,037	134,629

Table 95 - Loan Approval (Wilmington)

Loan Action	2018 %	2019 %	2020 %	2021 %	Total Period
Not approved (Denied, Withdrawn, or Incomplete)	33.25	30.60	29.02	29.03	6,861
Approved	66.75	69.40	70.98	70.97	16,081
Total N	3,083	4,559	7,323	7,977	22,942

APPENDIX J – SEX OF THE APPLICANT

Table 96 - Sex of the Applicant (Raleigh)

Sex	2018 %	2019 %	2020 %	2021 %	Total Period
Female	19.50	19.47	17.94	19.38	93,395
Male	33.74	33.73	34.19	32.22	164,468
Joint	30.19	30.48	31.18	31.17	152,104
Sex Not Specified	16.57	16.33	16.70	17.22	82,624
Total N	79,621	97,742	156,253	158,975	492,591

Table 97 - Sex of the Applicant (Durham)

Sex	2018 %	2019 %	2020 %	2021 %	Total Period
Female	23.92	23.22	21.90	23.21	39,365
Male	34.58	34.20	35.28	32.77	58,594
Joint	25.87	26.43	25.49	26.06	44,529
Sex Not Specified	15.63	16.14	17.32	17.96	29,199
Total N	27,540	36,383	52,246	55,518	171,687

Table 98 - Sex of the Applicant (Greensboro/High Point)

Sex	2018 %	2019 %	2020 %	2021 %	Total Period
Female	23.31	22.38	21.98	24.24	38,158
Male	31.19	31.77	32.00	30.46	51,904
Joint	30.31	30.77	30.11	30.44	50,367
Sex Not Specified	15.19	15.09	15.92	14.86	25,320
Total N	30,149	34,796	48,177	52,627	165,749

Table 99 - Sex of the Applicant (Winston-Salem)

Sex	2018 %	2019 %	2020 %	2021 %	Total Period
Female	22.82	21.94	20.95	22.84	33,704
Male	32.80	33.86	34.30	32.35	50,767
Joint	30.00	30.13	29.75	29.84	45,584
Sex Not Specified	14.38	14.07	15.00	14.96	22,376
Total N	28,237	31,643	43,439	49,112	152,431

Table 100 - Sex of the Applicant (Charlotte)

Sex	2018 %	2019 %	2020 %	2021 %	Total Period
Female	21.37	20.98	20.09	21.29	149,983
Male	32.73	33.16	33.14	33.32	238,022
Joint	30.00	30.39	31.33	29.27	217,375
Sex Not Specified	15.89	15.47	15.43	16.12	113,050
Total N	124,025	150,457	217,780	226,168	718,430

Table 101 - Sex of the Applicant (Asheville)

Sex	2018 %	2019 %	2020 %	2021 %	Total Period
Female	22.46	22.19	21.21	21.65	29,327
Male	26.44	25.81	26.19	26.40	35,302
Joint	38.77	39.13	39.19	37.81	52,060
Sex Not Specified	12.32	12.87	13.42	14.14	17,940
Total N	24,852	28,227	39,513	42,037	134,629

Table 102 - Sex of the Applicant (Wilmington)

Sex	2018 %	2019 %	2020 %	2021 %	Total Period
Female	13.43	12.15	11.17	12.34	2,770
Male	30.10	31.56	29.51	28.68	6,816
Joint	37.17	37.88	41.05	39.15	9,002
Sex Not Specified	19.30	18.40	18.27	19.83	4,354
Total N	3,083	4,559	7,323	7,977	22,942

APPENDIX K – SEX OF THE CO-APPLICANT

Table 103 - Sex of the Co-Applicant (Raleigh)

Sex	2018 %	2019 %	2020 %	2021 %	Total Period
Male	8.95	9.20	9.35	9.56	45,919
Female	26.47	26.07	26.29	24.20	126,109
Information not provided by applicant	3.67	3.87	4.22	4.37	20,253
Not applicable	6.97	7.26	6.58	6.55	33,346
No co-applicant	53.90	53.57	53.54	55.30	266,843
Co-applicant selected both male and female	0.03	0.02	0.02	0.03	121
Total N	79,621	97,742	156,253	158,975	492,591

Table 104 - Sex of the Co-Applicant (Durham)

Sex	2018 %	2019 %	2020 %	2021 %	Total Period
Male	10.23	10.41	11.01	10.76	18,330
Female	26.72	25.81	26.34	24.08	43,877
Information not provided by applicant	3.95	4.16	4.65	4.91	7,756
Not applicable	5.70	6.31	6.69	6.88	11,177
No co-applicant	53.38	53.27	51.30	53.36	90,506
Co-applicant selected both male and female	0.02	0.03	0.02	0.02	41
Total N	27,540	36,383	52,246	55,518	171,687

Table 105 - Sex of the Co-Applicant (Greensboro/High Point)

Sex	2018 %	2019 %	2020 %	2021 %	Total Period
Male	8.34	8.54	9.19	9.78	15,061
Female	24.68	24.74	24.34	22.47	39,600
Information not provided by applicant	3.09	2.99	3.96	3.77	5,863
Not applicable	6.36	7.11	5.88	4.83	9,767
No co-applicant	57.48	56.58	56.60	59.12	95,404
Co-applicant selected both male and female	0.06	0.03	0.03	0.02	54
Total N	30,149	34,796	48,177	52,627	165,749

Table 106 - Sex of the Co-Applicant (Winston-Salem)

Sex	2018 %	2019 %	2020 %	2021 %	Total Period
Male	8.69	9.34	10.11	10.03	14,723
Female	26.00	26.05	25.80	23.94	38,552
Information not provided by applicant	2.89	2.98	4.10	4.22	5,609
Not applicable	6.25	6.60	5.31	4.35	8,295
No co-applicant	56.14	55.03	54.65	57.45	85,222
Co-applicant selected both male and female	0.04	0.01	0.03	0.01	30
Total N	28,237	31,643	43,439	49,112	152,431

Table 107 - Sex of the Co-Applicant (Charlotte)

Sex	2018 %	2019 %	2020 %	2021 %	Total Period
Male	8.33	8.72	8.97	8.99	63,317
Female	23.43	23.30	23.93	21.90	165,756
Information not provided by applicant	3.36	3.40	3.92	3.87	26,577
Not applicable	6.10	6.52	5.13	5.22	40,343
No co-applicant	58.75	58.03	58.03	60.00	422,255
Co-applicant selected both male and female	0.04	0.02	0.02	0.02	182
Total N	124,025	150,457	217,780	226,168	718,430

Table 108 - Sex of the Co-Applicant (Asheville)

Sex	2018 %	2019 %	2020 %	2021 %	Total Period
Male	10.18	11.40	11.42	12.32	15,441
Female	31.27	30.57	30.31	28.23	40,244
Information not provided by applicant	2.98	3.47	4.32	4.32	5,240
Not applicable	4.63	5.01	4.32	4.86	6,314
No co-applicant	50.91	49.50	49.60	50.24	67,343
Co-applicant selected both male and female	0.04	0.05	0.03	0.03	47
Total N	24,852	28,227	39,513	42,037	134,629

Table 109 - Sex of the Co-Applicant (Wilmington)

Sex	2018 %	2019 %	2020 %	2021 %	Total Period
Male	8.76	8.80	10.53	10.97	2,317
Female	30.10	30.66	32.00	29.84	7,049
Information not provided by applicant	3.15	3.40	4.23	4.59	928
Not applicable	10.96	11.34	9.23	9.29	2,272
No co-applicant	47.03	45.80	44.00	45.31	10,374
Co-applicant selected both male and female	0.00	0.00	0.01	0.01	2
Total N	3,083	4,559	7,323	7,977	22,942

APPENDIX L – RACE/ETHNICITY RECODE OF THE APPLICANT

Table 110 - Race/Ethnicity Recode of the Applicant (Raleigh)

Race/Ethnicity Recode	2018 %	2019 %	2020 %	2021 %	Total Period
African American	10.72	10.42	8.79	9.83	48,099
American Indian	0.21	0.24	0.20	0.20	1,035
Asian	7.15	6.78	8.78	9.49	41,131
Hispanic	0.49	0.50	0.59	0.75	2,988
Multiracial Non-White	0.20	0.16	0.16	0.18	840
Multiracial White	1.48	1.66	1.64	1.74	8,136
Non-Hispanic White	58.02	57.63	56.46	53.01	275,032
Other/Free form Text	21.53	22.46	23.26	24.68	114,678
Pacific Islander	0.19	0.15	0.11	0.11	652
Total N	79,621	97,742	156,253	158,975	492,591

Table 111 - Race/Ethnicity Recode of the Applicant (Durham)

Race/Ethnicity Recode	2018 %	2019 %	2020 %	2021 %	Total Period
African American	14.07	14.41	11.80	12.45	22,189
American Indian	0.26	0.20	0.19	0.22	364
Asian	4.51	4.24	4.71	5.79	8,466
Hispanic	0.47	0.54	0.50	0.63	936
Multiracial Non-White	0.21	0.16	0.23	0.23	84365
Multiracial White	1.82	1.93	2.17	2.04	3,468
Non-Hispanic White	58.14	56.72	56.80	53.65	96,108
Other/Free form Text	20.37	21.67	23.48	24.90	39,588
Pacific Islander	0.15	0.14	0.11	0.09	203
Total N	27,540	36,383	52,246	55,518	171,687

Table 112 - Race/Ethnicity Recode of the Applicant (Greensboro/High Point)

Race/Ethnicity Recode	2018 %	2019 %	2020 %	2021 %	Total Period
African American	15.08	15.42	14.03	16.44	25,323
American Indian	0.22	0.27	0.26	0.32	453
Asian	3.38	3.18	3.36	3.76	5,724
Hispanic	0.45	0.49	0.47	0.63	867
Multiracial Non-White	0.21	0.16	0.17	0.28	347
Multiracial White	1.06	1.21	1.28	1.30	2,041
Non-Hispanic White	60.43	59.48	59.30	55.94	96,924
Other/Free form Text	19.03	19.66	21.02	21.20	33,860
Pacific Islander	0.14	0.13	0.13	0.12	210
Total N	30,149	34,796	48,177	52,627	165,749

Table 113 - Race/Ethnicity Recode of the Applicant (Winston-Salem)

Race/Ethnicity Recode	2018 %	2019 %	2020 %	2021 %	Total Period
African American	9.39	9.25	8.18	9.72	13,902
American Indian	0.29	0.21	0.24	0.29	396
Asian	1.57	1.42	1.87	1.98	2,674
Hispanic	0.39	0.30	0.35	0.50	606
Multiracial Non-White	0.12	0.13	0.13	0.13	192
Multiracial White	1.02	1.23	1.27	1.41	1,922
Non-Hispanic White	69.24	69.13	68.18	65.16	103,042
Other/Free form Text	17.83	18.16	19.67	20.70	29,494
Pacific Islander	0.14	0.17	0.13	0.11	203
Total N	28,237	31,643	43,439	49,112	152,431

Table 114 - Race/Ethnicity Recode of the Applicant (Charlotte)

Race/Ethnicity Recode	2018 %	2019 %	2020 %	2021 %	Total Period
African American	12.81	12.26	10.83	12.00	85,071
American Indian	0.24	0.20	0.21	0.23	1,589
Asian	4.03	4.15	5.15	5.84	35,664
Hispanic	0.55	0.51	0.60	0.69	4,315
Multiracial Non-White	0.17	0.16	0.15	0.19	1,188
Multiracial White	1.20	1.28	1.32	1.28	9,200
Non-Hispanic White	59.95	59.66	58.92	55.37	417,668

Other/Free form Text	20.88	21.62	22.69	24.26	162,711
Pacific Islander	0.17	0.14	0.13	0.14	1,024
Total N	124,025	150,457	217,780	226,168	718,430

Table 115 - Race/Ethnicity Recode of the Applicant (Asheville)

Race/Ethnicity Recode	2018 %	2019 %	2020 %	2021 %	Total Period
African American	1.61	1.93	1.43	1.60	2,186
American Indian	0.22	0.23	0.23	0.30	336
Asian	0.89	0.77	0.90	0.84	1,149
Hispanic	0.16	0.12	0.13	0.16	192
Multiracial Non-White	0.05	0.05	0.03	0.06	62
Multiracial White	1.95	1.16	1.22	1.36	1,869
Non-Hispanic White	79.52	78.78	78.09	75.90	104,763
Other/Free form Text	15.50	16.87	17.88	19.72	23,970
Pacific Islander	0.10	0.07	0.08	0.08	102
Total N	24,852	28,227	39,513	42,037	134,629

Table 116 - Race/Ethnicity Recode of the Applicant (Wilmington)

Race/Ethnicity Recode	2018 %	2019 %	2020 %	2021 %	Total Period
African American	4.87	4.56	3.74	3.87	941
American Indian	0.23	0.46	0.33	0.33	78
Asian	0.62	0.75	0.81	0.69	167
Hispanic	0.26	0.09	0.20	0.18	41
Multiracial Non-White	0.16	0.04	0.16	0.08	25
Multiracial White	1.23	1.27	1.82	1.94	384
Non-Hispanic White	69.67	68.74	67.95	66.52	15,564
Other/Free form Text	22.90	23.97	24.89	26.33	5,722
Pacific Islander	0.06	0.11	0.10	0.08	20
Total N	*3,083*	*4,559*	*7,323*	*7,977*	*22,942*

APPENDIX M – COMMUNITY MEASURES

Table 117 - Community Measures in Mean (Raleigh)

	2018	2019	2020	2021	Total Period
Population	7049	7065	7004	7087	7050
Percent Minority (%)	34.38	34.13	33.23	33.96	33.83
Tract to MSA Income Percent (%)	109.81	110.66	114.80	111.59	112.13
Owner Occupied Units	1838	1844	1837	1852	1843
Single Family (1-4)	0.998	0.999	0.999	0.999	0.999
Total N	79,621	97,742	156,253	158,975	492,591

Table 118 - Community Measures in Mean (Durham)

	2018	2019	2020	2021	Total Period
Population	5809	5826	5817	5815	5817
Percent Minority (%)	41.13	41.08	39.67	40.20	40.38
Tract to MSA Income Percent (%)	115.33	116.75	119.88	117.92	117.86
Owner Occupied Units	1587	1586	1593	1585	1588
Single Family (1-4)	0.998	0.996	0.997	0.998	0.998
Total N	27,540	36,383	52,246	55,518	171,687

Table 119 - Community Measures in Mean (Greensboro/High Point)

	2018	2019	2020	2021	Total Period
Population	5320	5336	5340	5318	5329
Percent Minority (%)	33.17	32.87	31.56	32.89	32.55
Tract to MSA Income Percent (%)	119.87	120.71	124.81	121.31	121.94
Owner Occupied Units	1482	1491	1506	1482	1491
Single Family (1-4)	0.997	0.997	0.997	0.997	0.997
Total N	30,149	34,796	48,177	52,627	165,749

Table 120 - Community Measures in Mean (Winston-Salem)

	2018	2019	2020	2021	Total Period
Population	4919	4919	4920	4924	4921
Percent Minority (%)	24.75	24.53	23.31	23.77	23.98
Tract to MSA Income Percent (%)	114.68	115.43	118.76	116.23	116.50
Owner Occupied Units	1436	1446	1460	1452	1450
Single Family (1-4)	0.996	0.997	0.997	0.996	0.997
Total N	28,237	31,643	43,439	49,112	152,431

Table 121 - Community Measures in Mean (Charlotte)

	2018	2019	2020	2021	Total Period
Population	5004	5018	5046	5025	5026
Percent Minority (%)	34.04	33.50	32.11	32.81	32.96
Tract to MSA Income Percent (%)	121.91	124.97	129.66	125.85	126.14
Owner Occupied Units	1303	1314	1333	1321	1320
Single Family (1-4)	0.998	0.998	0.999	0.998	0.999
Total N	124,025	150,457	217,780	226,168	718,430

Table 122 - Community Measures in Mean (Asheville)

	2018	2019	2020	2021	Total Period
Population	4812	4805	4786	4786	4795
Percent Minority (%)	13.54	13.48	13.23	13.62	13.46
Tract to MSA Income Percent (%)	103.60	103.79	105.02	103.73	104.10
Owner Occupied Units	1391	1388	1387	1379	1385
Single Family (1-4)	0.997	0.998	0.998	0.999	0.998
Total N	24,852	28,227	39,513	42,037	134,629

Table 123 - Community Measures in Mean (Wilmington)

	2018	2019	2020	2021	Total Period
Population	3915	4263	4116	4060	4099
Percent Minority (%)	15.19	15.95	15.08	14.77	15.16
Tract to MSA Income Percent (%)	104.64	105.93	108.39	107.74	107.17
Owner Occupied Units	1210	1327	1286	1262	1276
Single Family (1-4)	0.999	0.999	0.999	0.999	0.999
Total N	3,083	4,559	7,323	7,977	22,942

APPENDIX N – REASONS FOR DENIAL

Table 124 - Reasons for Denial (Raleigh)

Loan Action	2018 %	2019 %	2020 %	2021 %	Total Period
Debt-to-income ratio	28.28	26.06	27.24	28.35	14,605
Employment history	1.05	1.00	1.48	1.38	665
Credit history	27.94	24.41	23.27	21.77	12,802
Collateral	14.28	11.96	10.25	9.41	5,982
Insufficient cash (down payment, closing costs)	2.67	2.23	2.18	1.82	1,166
Unverifiable information	3.83	3.83	4.45	4.85	2,277
Credit application incomplete	9.79	9.87	17.82	18.30	7,653
Mortgage insurance denied	0.01	0.02	0.03	0.01	10
Other	7.54	9.06	10.75	11.32	5,233
Not Applicable	4.61	11.56	2.53	2.78	2,767
Total N	10,811	12,853	14,837	14,659	53,160

Table 125 - Reasons for Denial (Durham)

Loan Action	2018 %	2019 %	2020 %	2021 %	Total Period
Debt-to-income ratio	28.56	27.48	29.79	29.20	5,685
Employment history	0.97	1.03	1.56	1.37	247
Credit history	27.31	24.26	22.02	21.05	4,628
Collateral	14.84	12.34	11.41	11.66	2,452
Insufficient cash (down payment, closing costs)	3.55	3.13	2.11	1.77	510
Unverifiable information	3.91	3.48	4.44	5.73	873
Credit application incomplete	10.40	9.02	15.82	16.26	2,574
Mortgage insurance denied	0.08	0.02	0.00	0.08	8
Other	7.74	9.32	11.13	10.55	1,940
Not Applicable	2.63	9.92	1.73	2.33	846
Total N	3,914	5,342	5,267	5,240	19,763

Table 126 - Reasons for Denial (Greensboro/High Point)

Loan Action	2018 %	2019 %	2020 %	2021 %	Total Period
Debt-to-income ratio	24.34	22.98	25.99	26.35	6,372
Employment history	1.13	1.11	1.73	1.26	335
Credit history	31.98	32.07	29.95	28.49	7,781
Collateral	14.90	12.99	10.93	10.65	3,123
Insufficient cash (down payment, closing costs)	3.52	2.53	2.26	1.72	627
Unverifiable information	3.62	3.89	5.96	7.06	1,335
Credit application incomplete	7.74	7.22	12.08	13.35	2,621
Mortgage insurance denied	0.03	0.02	0.02	0.03	6
Other	8.02	7.25	8.09	8.49	2,036
Not Applicable	4.71	9.96	2.99	2.60	1,262
Total N	5,772	6,137	6,548	7,041	25,498

Table 127 - Reasons for Denial (Winston-Salem)

Loan Action	2018 %	2019 %	2020 %	2021 %	Total Period
Debt-to-income ratio	23.39	22.17	25.93	24.73	5,282
Employment history	1.14	0.93	2.00	1.31	297
Credit history	32.99	30.65	27.19	27.42	6,431
Collateral	17.60	15.15	13.11	12.88	3,182
Insufficient cash (down payment, closing costs)	2.80	2.44	1.92	1.67	475
Unverifiable information	3.59	4.45	6.02	6.98	1,177
Credit application incomplete	7.70	8.36	12.64	13.92	2,382
Mortgage insurance denied	0.06	0.00	0.07	0.02	8
Other	7.49	7.91	9.46	9.33	1,886
Not Applicable	3.23	7.93	1.65	1.75	770
Total N	4,925	5,168	5,561	6,236	21,890

Table 128 - Reasons for Denial (Charlotte)

Loan Action	2018 %	2019 %	2020 %	2021 %	Total Period
Debt-to-income ratio	27.16	25.06	25.73	26.63	24,199
Employment history	1.28	0.98	1.76	1.38	1,256
Credit history	32.39	27.92	25.57	24.98	25,480
Collateral	14.65	13.10	11.25	10.33	11,320
Insufficient cash (down payment, closing costs)	2.52	2.15	2.14	1.77	1,972
Unverifiable information	3.79	3.78	4.96	5.68	4,261
Credit application incomplete	8.97	9.04	16.88	17.36	12,316
Mortgage insurance denied	0.11	0.01	0.01	0.05	40
Other	7.87	7.25	9.34	10.08	8,050
Not Applicable	1.28	10.72	2.36	1.72	3,789
Total N	20,132	23,655	24,179	24,717	92,683

Table 129 - Reasons for Denial (Asheville)

Loan Action	2018 %	2019 %	2020 %	2021 %	Total Period
Debt-to-income ratio	32.23	30.80	30.31	31.15	5,754
Employment history	1.36	1.22	1.47	1.36	251
Credit history	26.10	24.45	23.36	22.73	4,455
Collateral	15.20	14.78	12.58	12.54	2,533
Insufficient cash (down payment, closing costs)	2.23	2.14	1.80	1.40	346
Unverifiable information	4.11	4.58	7.36	8.65	1,168
Credit application incomplete	7.14	8.40	13.77	12.84	1,986
Mortgage insurance denied	0.09	0.02	0.00	0.02	406
Other	7.56	7.27	7.92	8.37	1,445
Not Applicable	3.97	6.33	1.43	0.94	561
Total N	4,257	4,344	4,889	5,015	18,505

Table 130 - Reasons for Denial (Wilmington)

Loan Action	2018 %	2019 %	2020 %	2021 %	Total Period
Debt-to-income ratio	23.23	22.43 0.72	25.25	26.01	684
Employment history	0.00	1.22	1.00	1.10	22
Credit history	26.26	24.46	24.12	20.25	657
Collateral	21.62	13.60	12.38	13.13	407
Insufficient cash (down payment, closing costs)	3.23	2.89	2.62	2.45	77
Unverifiable information	3.03	3.76	3.12	7.12	124
Credit application incomplete	7.07	8.54	15.50	16.20	350
Mortgage insurance denied	0.20	0.00	0.00	0.00	1
Other	13.74	11.43	12.12	11.29	336
Not Applicable	1.62	12.16	3.88	2.45	143
Total N	495	691	800	815	2,801

APPENDIX O – APPROVAL BY LOAN TYPE

Table 131 - Approval by Loan Type (Raleigh)

Loan Type	2018 %	2019 %	2020 %	2021 %	Total Period %
Conventional	71.53	72.51	72.32	73.29	72.56
FHA	66.40	67.83	69.95	69.95	69.43
VA	67.41	69.59	72.62	70.51	70.57
FSA/RHS	84.43	80.19	84.47	83.67	83.41

Table 132 - Approval by Loan Type (Durham)

Loan Type	2018 %	2019 %	2020 %	2021 %	Total Period %
Conventional	71.54	71.32	72.18	73.52	72.35
FHA	61.57	64.12	68.58	66.63	65.55
VA	62.85	64.68	71.29	69.44	68.07
FSA/RHS	80.00	82.24	81.56	80.88	81.32

Table 133 - Approval by Loan Type (Greensboro/High Point)

Loan Type	2018 %	2019 %	2020 %	2021 %	Total Period %
Conventional	66.31	67.40	66.88	68.02	67.26
FHA	66.35	68.54	70.82	68.05	68.57
VA	61.59	66.32	69.86	66.83	66.82
FSA/RHS	84.58	83.23	80.40	84.33	82.92

Table 134 - Approval by Loan Type (Winston-Salem)

Loan Type	2018 %	2019 %	2020 %	2021 %	Total Period %
Conventional	67.40	69.65	68.19	68.98	68.61
FHA	67.72	68.15	71.21	69.47	69.25
VA	61.43	66.09	70.11	66.94	66.78
FSA/RHS	81.79	79.44	83.68	79.56	81.38

Table 135 - Approval by Loan Type (Charlotte)

Loan Type	2018 %	2019 %	2020 %	2021 %	Total Period %
Conventional	67.74	69.18	70.41	71.31	70.01
FHA	65.28	67.30	69.39	67.84	67.60
VA	63.22	66.86	70.63	68.60	68.01
FSA/RHS	78.95	76.69	79.77	79.39	78.85

Table 136 - Approval by Loan Type (Asheville)

Loan Type	2018 %	2019 %	2020 %	2021 %	Total Period %
Conventional	67.71	69.33	68.70	69.17	68.80
FHA	61.71	64.15	67.68	65.36	64.82
VA	61.67	62.24	68.63	64.96	64.98
FSA/RHS	81.75	76.77	81.45	81.36	80.39

Table 137 - Approval by Loan Type (Wilmington)

Loan Type	2018 %	2019 %	2020 %	2021 %	Total Period %
Conventional	63.44	64.90	66.95	68.51	66.66
FHA	67.90	72.42	72.39	70.73	71.12
VA	74.78	75.35	77.78	76.31	76.46
FSA/RHS	75.44	89.06	81.63	86.32	83.44

APPENDIX P – APPROVAL RATES BY SEX

Table 138 - Approval Rates by Sex (Raleigh)

Sex	2018 %		2019 %		2020 %		2021 %		Total Period	
	Approved	Not Approved	Approved	Not Approved	Approved	Not Approved	Approved	Not Approved	Approved	Not Approved
Female	65.94	34.06	67.30	32.70	68.33	31.67	69.30	30.70	63,547	29,848
Male	65.28	34.72	67.05	32.95	68.18	31.82	68.40	31.60	102,780	49,324
Joint	73.75	26.25	74.05	25.95	74.14	25.86	75.65	24.35	122,573	41,895

Table 139 - Approval Rates by Sex (Durham)

Sex	2018 %		2019 %		2020 %		2021 %		Total Period	
	Approved	Not Approved	Approved	Not Approved	Approved	Not Approved	Approved	Not Approved	Approved	Not Approved
Female	65.92	34.08	65.45	34.55	67.61	32.39	68.78	31.22	26,472	12,893
Male	63.82	36.18	64.05	35.95	66.20	33.80	67.30	32.70	29,260	15,269
Joint	73.89	26.11	73.95	26.05	74.63	25.37	76.51	23.49	43,916	14,678

Table 140 - Approval Rates by Sex (Greensboro/High Point)

Sex	2018 %		2019 %		2020 %		2021 %		Total Period	
	Approved	Not Approved	Approved	Not Approved	Approved	Not Approved	Approved	Not Approved	Approved	Not Approved
Female	61.33	38.67	63.52	36.48	63.37	36.63	64.89	35.11	24,243	13,915
Male	61.07	38.93	61.86	38.14	63.36	36.64	64.39	35.61	31,708	18,659
Joint	68.48	31.52	70.19	29.81	70.19	29.81	71.51	28.49	36,482	15,422

Table 141 - Approval Rates by Sex (Winston Salem)

Sex	2018 %		2019 %		2020 %		2021 %		Total Period	
	Approved	Not Approved	Approved	Not Approved	Approved	Not Approved	Approved	Not Approved	Approved	Not Approved
Female	62.41	37.59	65.21	34.79	65.33	34.67	66.14	33.86	21,914	11,790
Male	62.54	37.46	64.53	35.47	64.58	35.42	64.89	35.11	29,306	16,278
Joint	69.23	30.77	71.04	28.96	71.40	28.60	72.21	27.79	36,137	14,630

Table 142 - Approval Rates by Sex (Charlotte)

Sex	2018 %		2019 %		2020 %		2021 %		Total Period	
	Approved	Not Approved	Approved	Not Approved	Approved	Not Approved	Approved	Not Approved	Approved	Not Approved
Female	62.12	37.88	64.21	35.79	66.98	33.02	67.32	32.68	98,459	51,524
Male	62.35	37.65	64.27	35.73	67.32	32.68	67.26	32.74	156,660	81,362
Joint	69.64	30.36	71.28	28.72	72.33	27.67	74.12	25.88	156,926	60,449

Table 143 - Approval Rates by Sex (Asheville)

Sex	2018 %		2019 %		2020 %		2021 %		Total Period	
	Approved	Not Approved	Approved	Not Approved	Approved	Not Approved	Approved	Not Approved	Approved	Not Approved
Female	62.56	37.44	64.92	35.08	65.49	34.51	64.97	35.03	18,960	10,367
Male	61.77	38.23	63.45	36.55	64.49	35.51	63.90	36.10	22,446	12,856
Joint	69.67	30.33	70.79	29.21	71.23	28.77	71.94	28.06	36,997	15,063

Table 144 - Approval Rates by Sex (Wilmington)

Sex	2018 %		2019 %		2020 %		2021 %		Total Period	
	Approved	Not Approved	Approved	Not Approved	Approved	Not Approved	Approved	Not Approved	Approved	Not Approved
Female	54.83	45.17	56.14	43.86	66.14	33.86	58.74	41.26	1,657	1,113
Male	59.38	40.62	65.60	34.40	65.34	34.66	65.95	34.05	4,416	2,400
Joint	67.28	32.72	69.72	30.28	71.16	28.84	72.56	27.44	6,380	2,622